

FCE 201: Theory of Structures 1A

Class Notes

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Course Content

1. Introduction to statics.
2. Plane statics.
3. Structural forms and introduction of statically determinant structures.
4. Axial force structures - tension (cable) structures and compression (arch) structures.
5. Truss forms and analysis of trussels - method of joints, method of sections, and method of tension coefficients.
6. Bending structures - shear force and bending moment diagrams.

Introductory Notes

Fields of Civil Engineering

The field of Civil Engineering includes several branches:

- Structural Engineering
- Water Resources Engineering
- Public Health Engineering
- Transportation Engineering
- Geotechnical Engineering

Structural Engineering Units

Structural Engineering components of civil engineering will comprise of the following courses

- Theory of Structures (2nd, 3rd, and 4th Years – 6 Course Units)
- Strength of Materials (2nd and 3rd Year – 4 course units)
- Civil Engineering Materials (2nd and 3rd Year – 4 course units)
- Engineering Mathematics (2nd and 3rd Years – 2 course units)
- Structural Design (4th and 5th Year – 4 course units)

2 STRUCTURES

2.1 Structures Definition

A structure can be defined as anything that either supports or resists forces of some kind, or acts as a protective cover for something or someone, or does both.

2.2 Groups of Structures

There are two main groups of structures:

2.2.1 Natural Structures

These are structures which occur in nature, eg animal skeletons, tree trunks, spider webs, etc

2.2.2 Artificial Structures

These are structures which are designed and constructed by man. They include buildings, bridges, dams, tunnels, silos, stadia, water tanks, etc (These are covered in civil engineering and are static). Other examples include wheelbarrows, bicycles, motor vehicles, ships, aircraft, etc. (These are dynamic and are covered under mechanical engineering)

2.3 Structural Forms

Structures may be divided into two categories depending on their structural form

1. Skeletal structures –

Structures which are made up of distinct, ideally linear members, Examples of skeletal structures include all frameworks (trusses, grills, frames and arches). Some of these structures will be dealt with in this unit FCE 201

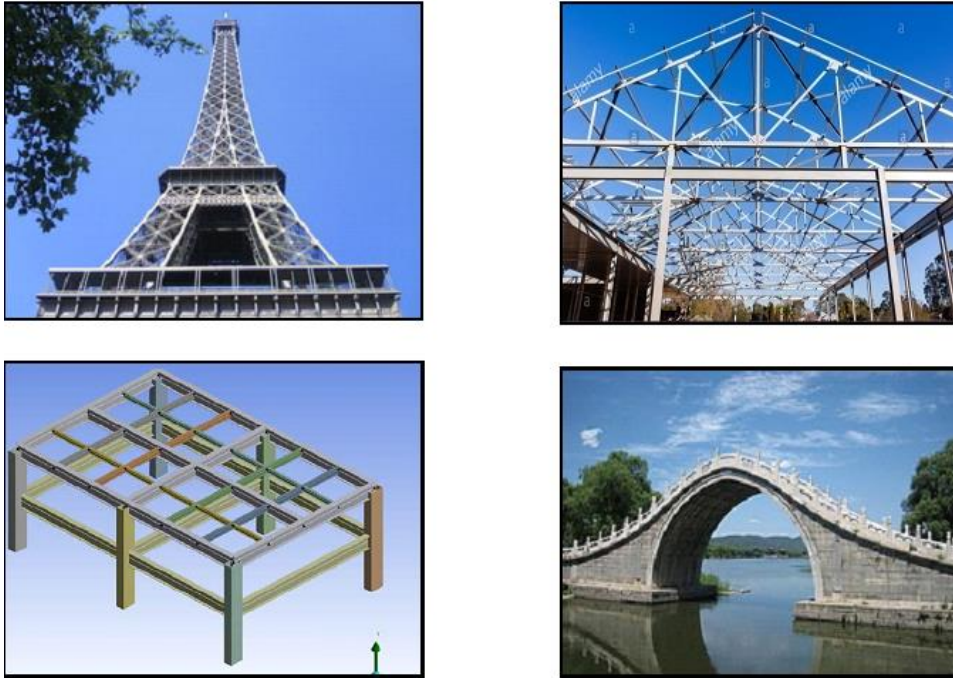


Figure 2-1: Skeletal structures, clockwise from top left; frame, grill, truss and arch structures

2. Continuum structures –

These are structures which are made up of members which can be defined in at least two dimensions and are continuous over space. Examples include floor slabs, walls, dams, roof shells, etc

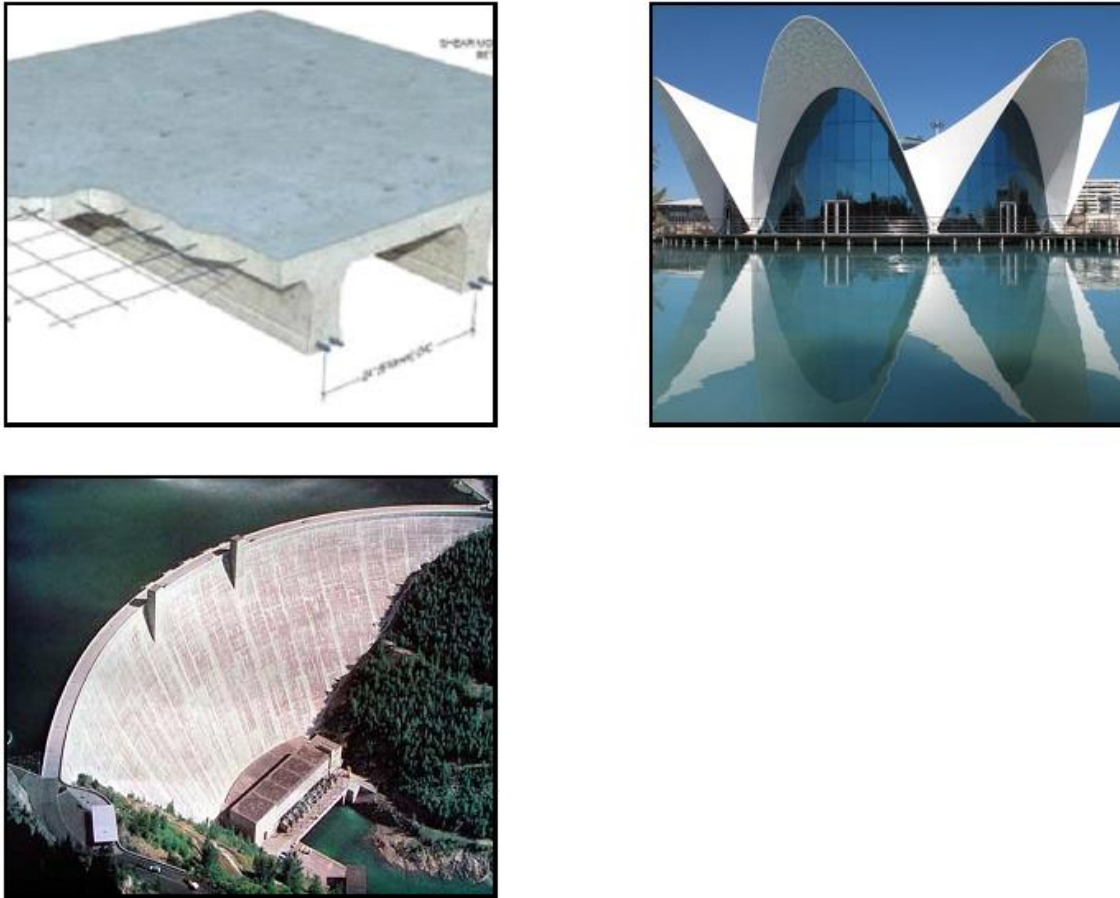


Figure 2-2: Continuum structures

2.4 Structural Functional Units

A structure is considered to be composed of structural units as shown below:

- A structure is made up of several structural systems working together
- Each structural system is made up of several members inter-connected and functioning in such a manner as to give complete structural independence
- Each member is made up of a series of cross sections from one end to the other
- Each cross-section is seen to be suffering a specific level of the load effect

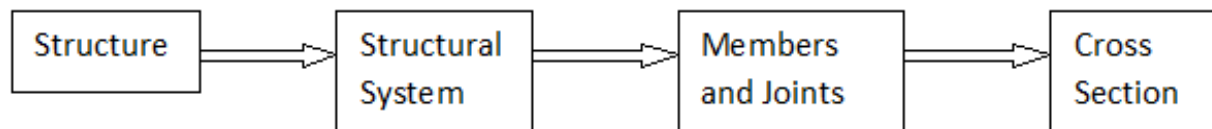


Figure 2-3: Structure - Components

2.5 Structural Analysis

When a load is applied to a structure or a structural element, the structure or structural element will respond by developing stresses (pressures) in its material. As a consequence of these stresses, the material will develop strains (deformations) resulting in the global deformation of the structure or structural element¹.

The aim of structural analysis is to predict (ie determine in advance) the internal stresses that will be induced in the material of a structure due to a given set of loads, and deformations that the structure will have to suffer. The internal stresses are called member forces, while the deformations are referred to as displacements.

2.6 Structural Design

Structural engineering refers to all that engineering work (modeling, analysis, specifications, detailing) which results in creating on paper how what is proposed must be constructed on site, so that its structural behavior is acceptable

Structural behavior is considered acceptable if the structure is safe, stable and serviceable throughout its design life

2.7 Conclusion

Figure 2-4 summarises the whole process of structural design

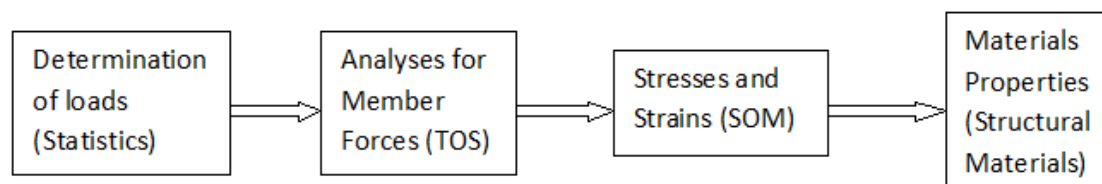


Figure 2-4: Structural Design Stages

3 CONCEPTS AND PRINCIPLES

3.1 A Structure

A structure is defined as an assembly of members joined together and supported in such a manner as to form a stable framework. The framework must be able to transmit the applied loads safely from

¹ It is important to note that it is physically impossible for stresses to exist in a material without corresponding strains, and vice versa. The two will always exist together

their points of application, via the joints, to the points of restraint in a stable manner and without excessive deformation

3.2 Structural Analysis

Refers to the techniques used in the prediction of the behavior of a structure when subjected to applied loads. Many theories have been formulated for carrying out these predictions. Theory of structures refers to these theories.

This course (FCE 201) will cover the determination of external forces induced at the supports of structures (referred to as support reactions), and internal forces induced in the members of the structures due to a given set of loads acting on a given structure (referred to as member forces).

3.3 Structural idealisation

Structural idealization is the simplification of an actual structural system to a model that can be analyzed with relative ease using mathematical and physical principles. Such simplification is only for the purpose of structural analysis.

This simplification is made by making appropriate assumptions regarding:

- The nature of the applied loads
- The nature and behavior of the members
- The nature and behavior of the joints
- The nature and behavior of the points of restraint

Structural idealization results in what is termed as idealized structure or structural model

3.3.1 Idealization of Applied Loads

Within the scope of this course, applied loads will be idealized into three categories:

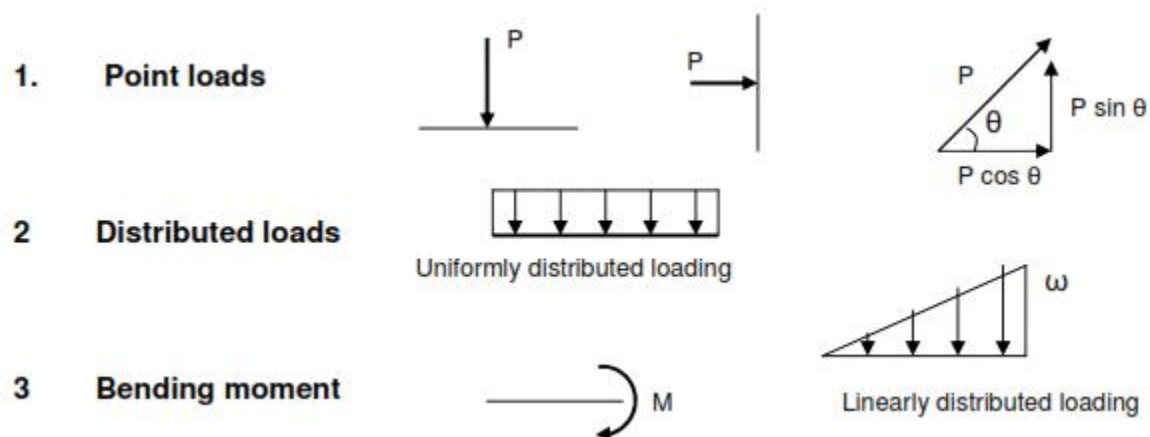


Figure 3-1: Idealization of applied loads

For purposes of analysis, all applied loads shall be considered with regard to the Cartesian coordinate system; prior knowledge of resolution of forces will come in handy

3.3.2 Idealization of Structural Members

In idealizing structural members, the following assumptions are made:

1. The member is prismatic, or is linearly tapered
2. The member is linear or curvilinear
3. The material is elastic, isotropic, and homogeneous
4. The material is not stressed beyond the elastic limit
5. The member may be represented by a line coinciding with its longitudinal axis
6. The self weight of the member is negligible

3.3.3 Idealization of Joints

Joints are idealized into two categories:

Pin Joint (Hinge)

A pin joint can resist force in the X and/or Y directions but cannot resist bending moment about the pin

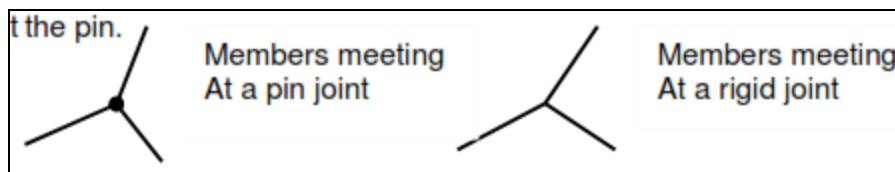


Figure 3-2: Joints idealization

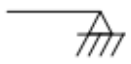
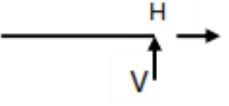
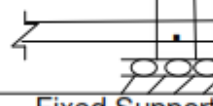
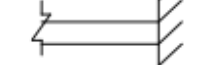
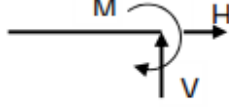
Rigid Joint

A rigid joint can resist force in the X and Y directions, and can resist bending moments about the joint

3.3.4 Idealization of Supports

There are three basic types of supports as shown in Table 3-1 below

Table 3-1: Basic Types of Supports

Real Model	Idealized Model	Free body diagram	Remarks
Pin Support 			Can resist a force in the x-direction Can resist a force in the y-direction Cannot resist moment about the z-axis
Roller Support 			Can resist a force in the y-direction Cannot resist a force in the x-direction Cannot resist moment about the z-axis
Fixed Support 			Can resist a force in the x-direction Can resist a force in the y-direction Can resist moment about the z-axis

3.3.5 Free body diagram

A “free body” is one that is imagined to be freed (ie totally disconnected) from all of its surrounding members, joints, or supports, whose presence is acknowledged only by the forces and moments they apply to the free body.

A free body may consist of part of a member, or a whole member, or several members, or the whole structure; but the free body must be able to sustain all the forces and moments acting on it in equilibrium, and in a stable manner

A sketch of a free body showing the body and all the forces and moments acting on it is referred to as a free body diagram

3.4 Principles

3.4.1 Principle of Static Equilibrium

The principle of static equilibrium states that, for a free body to remain static, the algebraic sum of all the resolved forces in any particular direction must equal zero, and the resultant moment must also equal zero

Consider the free body shown below (which is a 2D system)

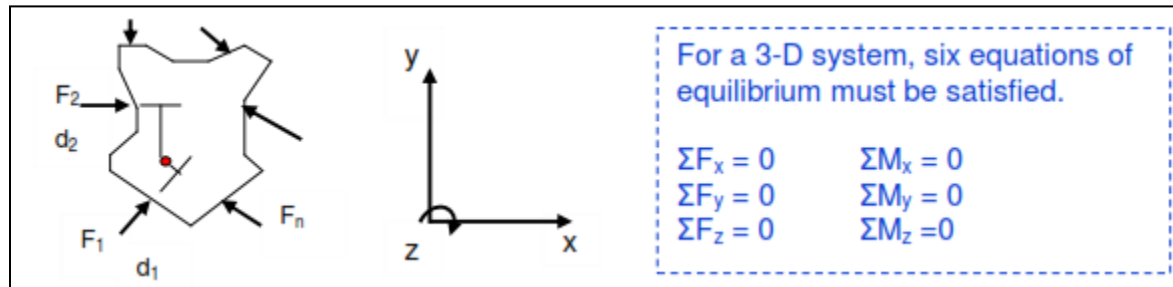


Figure 3-3: Free body diagram

The principle of static equilibrium requires that, for this free body to be static equilibrium, three equations must be satisfied:

$$\begin{aligned}\Sigma F_x &= F_1 \cos \theta_1 + F_2 \cos \theta_2 + \dots + F_n \cos \theta_n = 0 \\ \Sigma F_y &= F_1 \sin \theta_1 + F_2 \sin \theta_2 + \dots + F_n \sin \theta_n = 0 \\ \Sigma M_z &= F_1 d_1 + F_2 d_2 + \dots + F_n d_n = 0\end{aligned}$$

Where θ is the inclination of the i th force with respect to the x -axis

In the subsequent discussions the principle of static equilibrium will be used to determine the reaction components induced at the support of structures due to a given set of loads.

3.4.2 Principle of Compatibility

The principle of compatibility states that the strains distribution over the whole structure must be compatible with the shape of the deformed structure. In simpler terms, the principle states that the structural form must show continuity both before and after straining

3.4.3 Principle of Constitutivity

The principle of constitutivity states that the stresses must be related to the strains by means of the material constitutive properties. In simpler terms the principle states that the stresses are related to the strains by Hooke's law of linear elasticity Thus $\sigma = E\epsilon$

3.4.4 Principle of superposition

The principle of superposition states that the structural response due to a number of simultaneously applied loads is equal to the algebraic sum of the responses due to the individual loads each applied separately to the structure

3.5 Determination of Support Reactions

3.5.1 Procedure

The procedure for the determination of support reaction components is as follows:

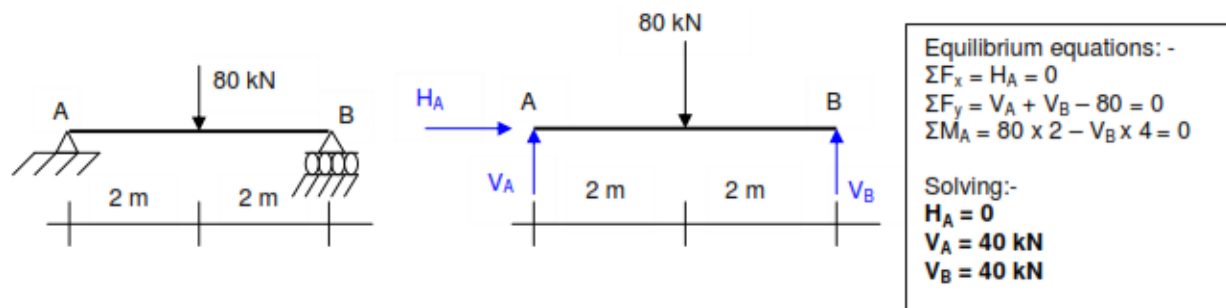
- Step 1: Isolate and sketch an appropriate free body diagram
All the external forces and moments acting on it must be shown
- Step 2: Set up the equations of static equilibrium
- Step 3: Solve for the unknown forces

3.5.2 Worked Examples

Determine the reactions at the supports of each of the beams shown in examples 1 to 7

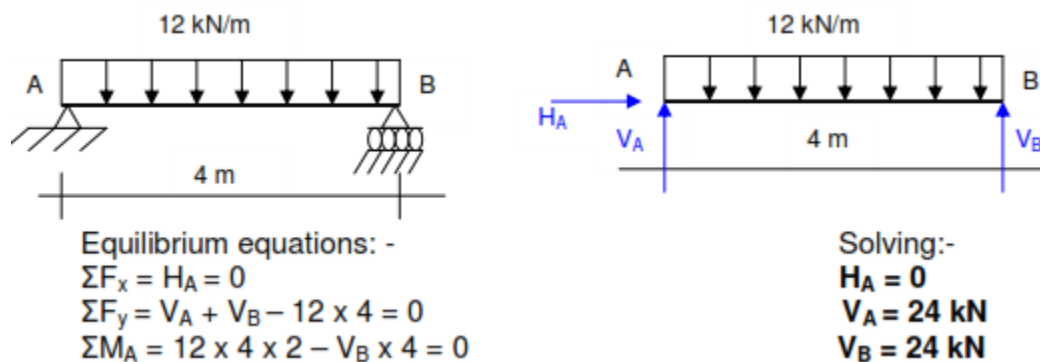
Example 1

A simply supported beam carrying a central point load



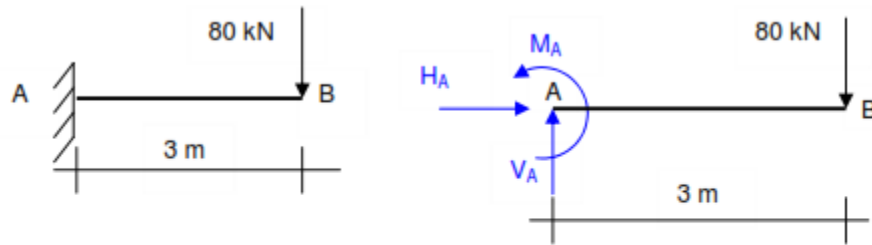
Example 2

A simply supported beam carrying a udl



Example 3

A cantilever carrying a point load at the free end



Equilibrium equations: -

$$\Sigma F_x = H_A = 0$$

$$\Sigma F_y = V_A - 80 = 0$$

$$\Sigma M_A = 80 \times 3 - M_A = 0$$

Solving:-

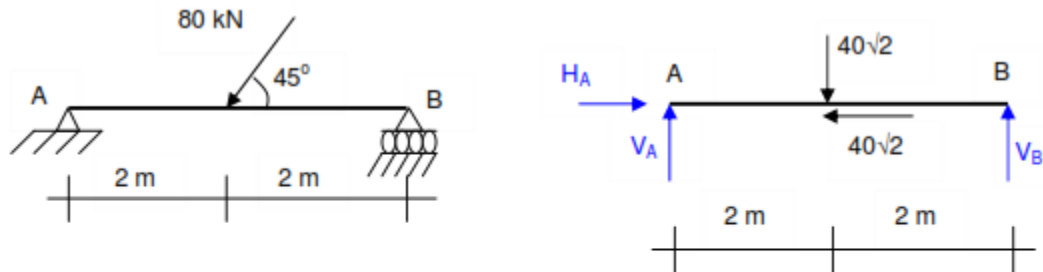
$$H_A = 0$$

$$V_A = 80 \text{ kN}$$

$$M_A = 240 \text{ kNm}$$

Example 4

A simply supported beam carrying an inclined point load



Equilibrium equations: -

$$\Sigma F_x = H_A - 40\sqrt{2} = 0$$

$$\Sigma F_y = V_A + V_B - 40\sqrt{2} = 0$$

$$\Sigma M_A = 40\sqrt{2} \times 2 - V_B \times 4 = 0$$

Solving:-

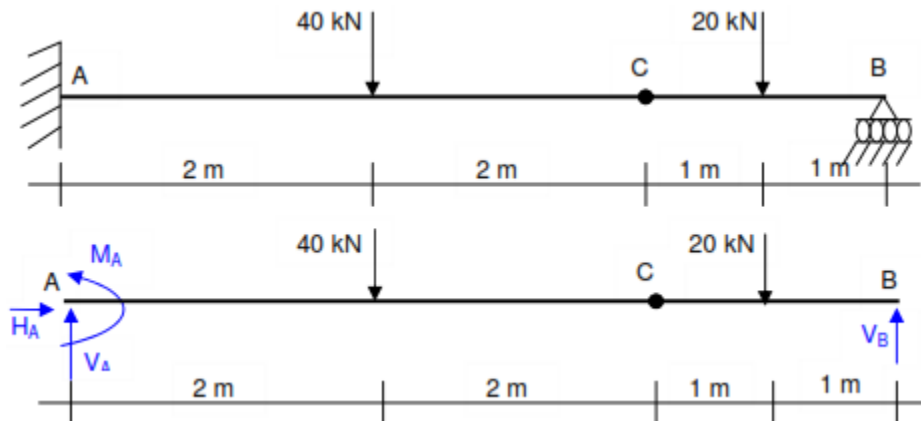
$$H_A = 40\sqrt{2} \text{ kN}$$

$$V_A = 20\sqrt{2} \text{ kN}$$

$$V_B = 20\sqrt{2} \text{ kN}$$

Example 5

A propped cantilever with an internal hinge



Equilibrium equations: -

$$\Sigma F_x = H_A = 0$$

$$\Sigma F_y = V_A + V_B - 40 - 20 = 0$$

$$\Sigma M_A = 40 \times 2 + 20 \times 5 - V_B \times 6 - M_A = 0$$

$$\Sigma M_{C(R)} = 20 \times 1 - V_B \times 2 = 0$$

Solving:-

$$H_A = 0$$

$$V_A = 50 \text{ kN}$$

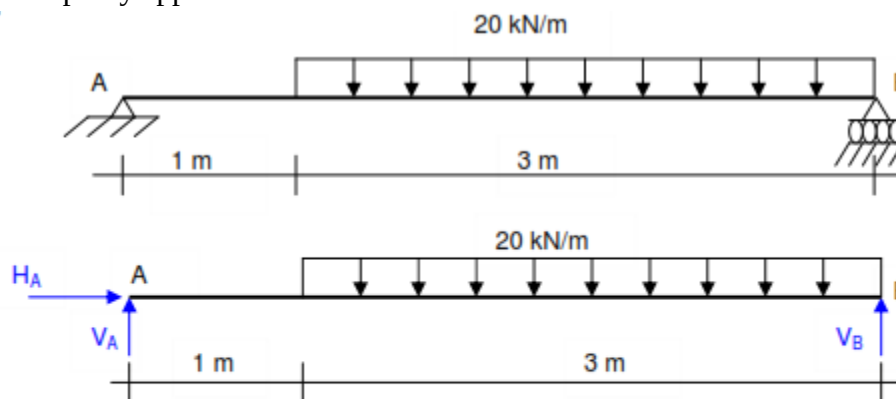
$$V_B = 10 \text{ kN}$$

$$M_A = 120 \text{ kNm}$$

NOTE: The above solution can alternatively be achieved by isolating two free bodies AC and CB.

Example 6

A beam with partly applied udl



Equilibrium equations: -

$$\Sigma F_x = H_A = 0$$

$$\Sigma F_y = V_A + V_B - 20 \times 3 = 0$$

$$\Sigma M_A = 20 \times 3 \times 2.5 - V_B \times 4 = 0$$

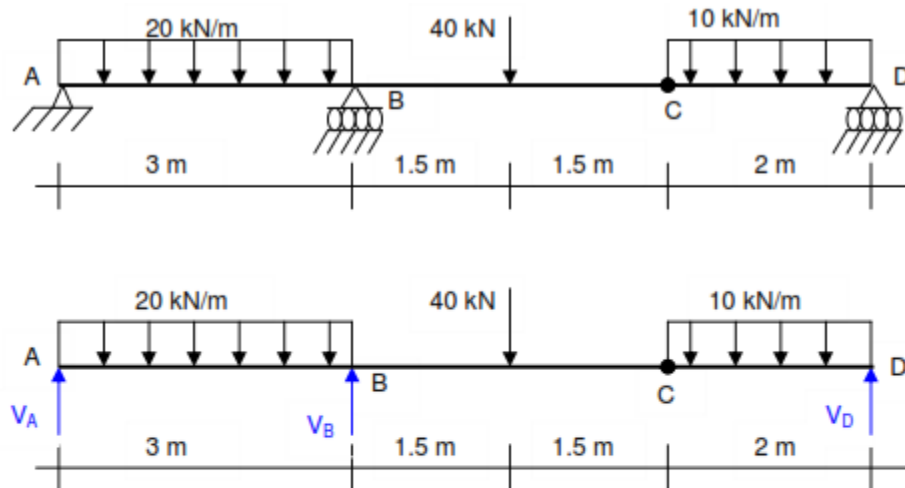
Solving:-

$$H_A = 0$$

$$V_A = 22.5 \text{ kN}$$

$$V_B = 37.5 \text{ kN}$$

Example 7



Equilibrium equations: -

$$\Sigma F_y = V_A + V_B + V_D - 20 \times 3 - 40 - 10 \times 2 = 0$$

$$\Sigma M_A = 20 \times 3 \times 1.5 + 40 \times 4.5 + 10 \times 2 \times 7 - 3 \times V_B - 8 \times V_D$$

$$\Sigma M_{C\odot} = 10 \times 2 \times 1 - 2 \times V_D = 0$$

Solving:-

$$V_A = 0$$

$$V_B = 110 \text{ kN}$$

$$V_D = 10 \text{ kN}$$

Note that so long as there are no applied loads with components in the x-direction, the horizontal reaction component H_A will turn out to be zero. Hence the equilibrium equation in the x-direction is not necessary.

3.6 Static Indeterminacy

3.6.1 Degree of static indeterminacy (dsi)

1. A structure is said to be statically indeterminate if all the external and internal force resultants can be determined by applying the equations of static equilibrium only
2. If the equations of static equilibrium are not sufficient to determine all external and internal force resultants, the structure is said to be statically indeterminate
3. The number of additional equations needed to solve for all the unknown forces is termed the degree of static indeterminacy (denoted by η). These additional equations are obtained by invoking the principle of compatibility and constitutivity
4. This course (FCE 201) will deal with statically determinate structures only. However, there is need to be able to distinguish between statically determinate and statically indeterminate structures. This can be done easily by evaluating the degree of static indeterminacy

3.6.2 Determination of Degree of Static Indeterminacy

For any given structure:

1. Isolate appropriate fbd(s) and identify all unknown external and internal forces, r
2. Determine the number of possible equations of equilibrium, e

3. Evaluate the dsi $\eta = r - e$

4. Make the decision as follows:

If $r > e \Rightarrow \eta$ is +ve $\Rightarrow$ the structure is statically indeterminate

If $r = e \Rightarrow \eta = 0 \Rightarrow$ the structure is statically determinate

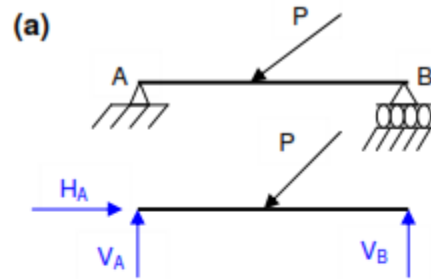
If $r < e \Rightarrow \eta$ is -ve $\Rightarrow$ the structure is **unstable!**

3.6.3 Geometric instability

1. A stable structure is one which is restrained (ie supported) in such a manner that it responds to any applied loading by developing stresses instantaneously, and hence the structure remains static and does not change its original geometry appreciably
2. An unstable structure is one which undergoes rigid body motion or changes its original geometry appreciably before developing any stresses. Such a structure is called a mechanism and is considered to suffer from geometric instability
3. The principle of static equilibrium and the concept of degree of static indeterminacy are not applicable to a mechanism. However, the structural engineer must be able to recognize geometric instability so as to provide additional restraints to stabilize the structure.

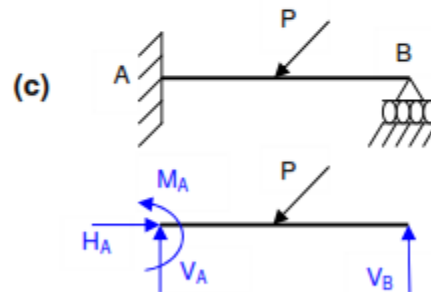
Example 8

Determine the degree of static determinacy for each of the structures shown below



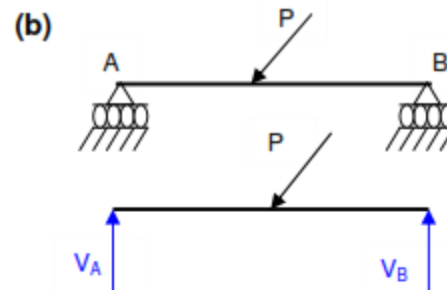
$$r = 3, e = 3, \text{ and } \eta = 0$$

Structure is **statically determinate**.



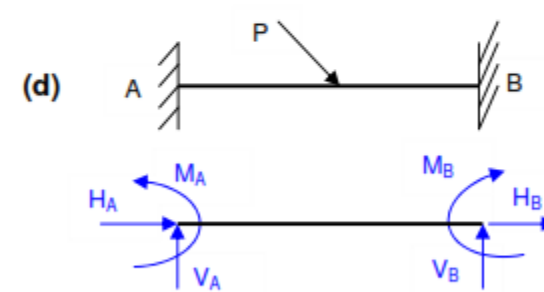
$$r = 4, e = 3 \text{ and } \eta = 1$$

Structure is **statically indeterminate to the first degree**



$$r = 2, e = 3 \text{ and } \eta = -1$$

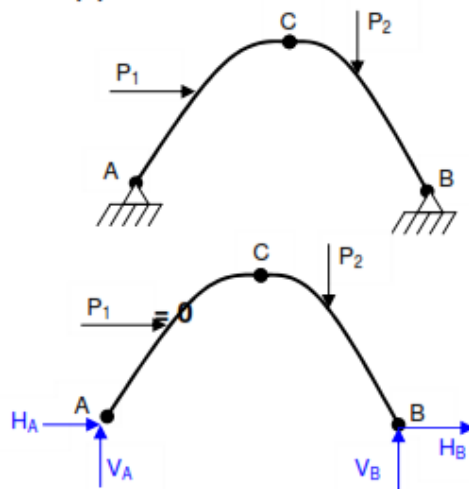
Structure is **unstable!**



$$r = 6, e = 3 \text{ and } \eta = 3$$

Structure is **statically indeterminate to the third degree**

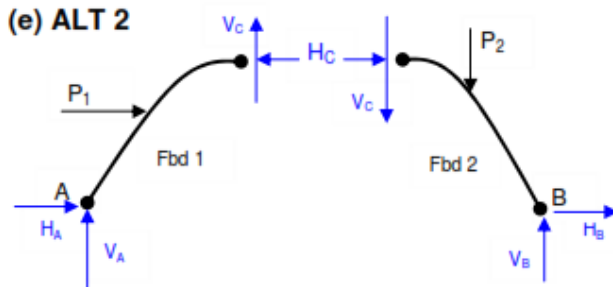
(e) ALT 1



$$r = 4; e = 3 + 1; \eta = 0$$

The structure is **statically determinate**

(e) ALT 2



$$\text{In this case: } r = 6; e = 3 + 3;$$

Note

1. The dsi is independent of the applied loads and/or the dimensions of the structure. The dsi depends only on the arrangement of supports, members and joints.
2. The choice of which fbd(s) to isolate is a matter of convenience other than principle.
3. When a "cut" is imagined through a member or joint The exact number of unknown internal forces and moments acting on that section must be revealed

3.7 Special Cases of Static Equilibrium

3.7.1 Case 1 (Case of only two forces)

If there are only two forces acting on a plane body which is in static equilibrium, then the forces must be equal in magnitude, opposite in direction and colinear

3.7.2 Case 2 (case of only three forces)

If there are only three forces acting on a plane body which is in static equilibrium, then the forces must be concurrent (ie they must all concur at some point). Consider the simply supported beam shown in Figure 3-4.

Since V_B must be vertical, and coincides with the line of action of P at the point O , then V_A and H_A must be such that their resultant R_A must pass through the point of concurrence, O .

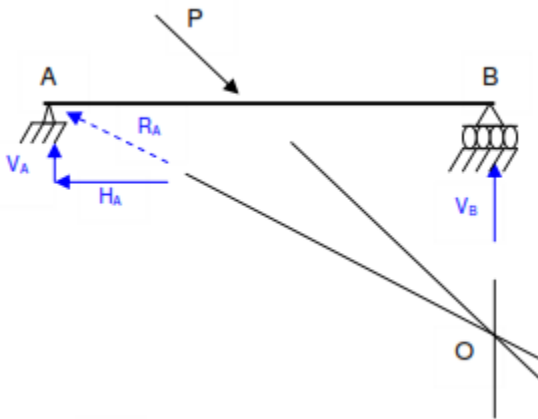


Figure 3-4: Concurrence of forces

3.7.3 Case 3 (Case of parallel forces)

A set of parallel forces acting on a plane body can be reduced to the resultant force or resultant couple. The algebraic sum of all the forces gives the resultant force:

$$R = \Sigma F_i$$

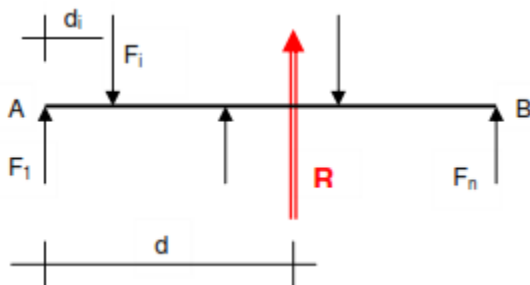


Figure 3-5: Parallel forces and their resultant

Sum of moments about, say point A, gives the distance d, which defines the point of action of R;

$$Rd = \Sigma F_i d_i$$

If $R = 0$, then either the system is in equilibrium or the system experiences as resultant couple. To determine which is which, sum moments about any point on the plane of the body, say point A

If $\Sigma M_A = 0$, then the system is in static equilibrium

If $\Sigma M_A = M$, then M is the resultant couple

3.8 Conclusion

The objective of FCE 201 is to enable the student to analyse statically determinate structures using the concepts and principles laid down in this chapter, together with the specific techniques that shall be learnt in subsequent chapters.

Statically determinate structures can be grouped into four categories. Each category shall constitute a chapter in this course.

The analyses will be limited to determining the support reaction components and internal member forces for any given set of loads.

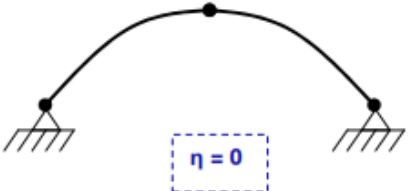
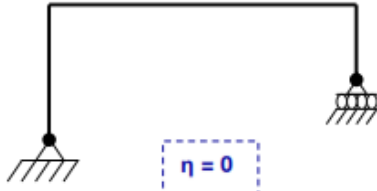
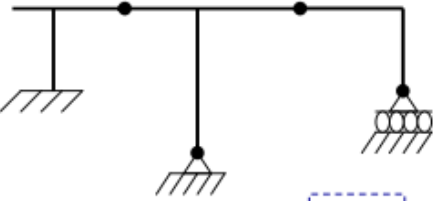
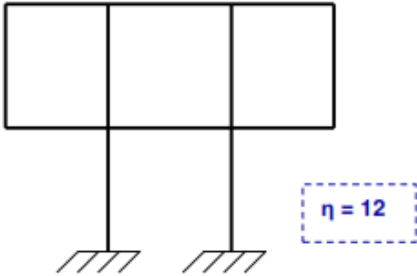
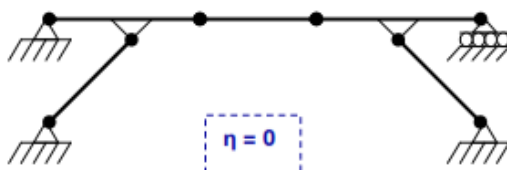
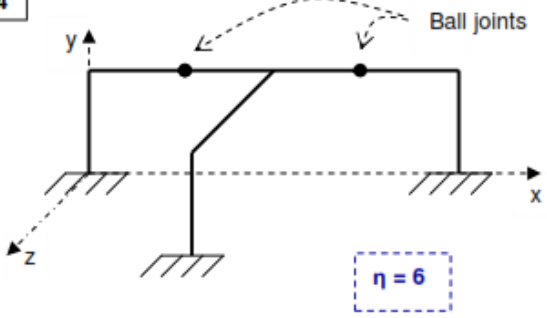
TUTORIAL 1: SUPPORT REACTIONS

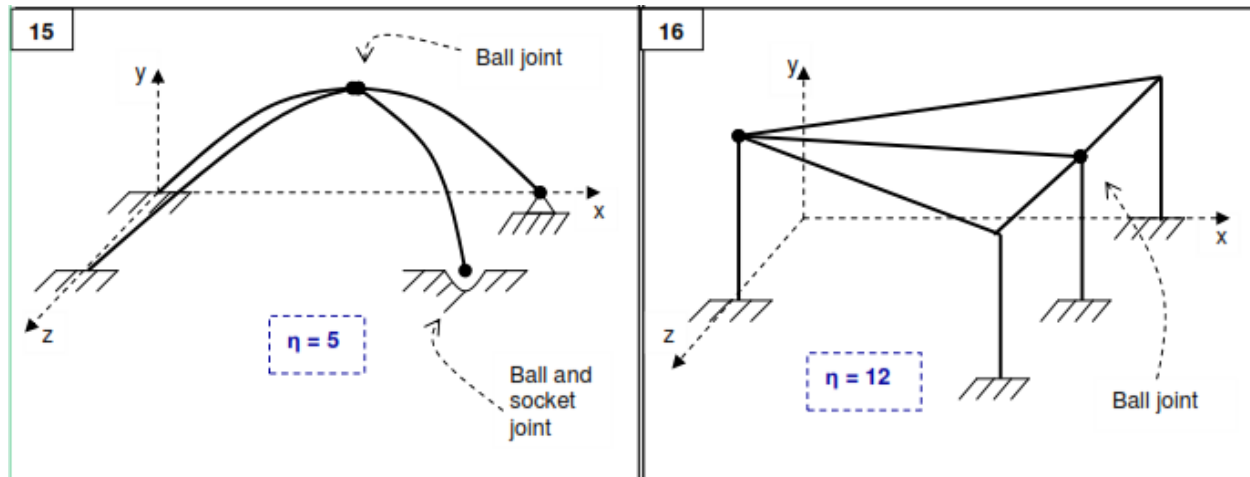
Determine the support reactions for each of the structures below

1 [Note: ABCD is a cable] $V_A = 10 \text{ kN}; H_A = -60 \text{ kN}; V_D = 10 \text{ kN}; H_D = 60 \text{ kN}$ $V_A = 1 \text{ kN}; H_A = 0.25 \text{ kN}; H_B = -0.25 \text{ kN}$	2	3 [Note: B is a rigid joint] $V_A = 6.75 \text{ kN}; H_A = -18 \text{ kN}$ $V_B = 29.25 \text{ kN}$
4 [Note: CDE is a cable] $V_A = 77.50 \text{ kN}; H_A = -103.92 \text{ kN}; V_B = 182.50 \text{ kN}$ $V_D = 60.00 \text{ kN}; H_D = 103.92 \text{ kN}$	5 $V_A = 42 \text{ kN}; H_A = 36 \text{ kN}$ $V_B = 198 \text{ kN}; V_D = 24 \text{ kN}$	
6 $V_A = 27 \text{ kN}; H_A = 12 \text{ kN}$ $V_E = 93 \text{ kN}; H_E = -42 \text{ kN}$	7 $V_A = 58.5 \text{ kN}$ $H_A = 34.5 \text{ kN}$ $V_B = 79.5 \text{ kN}$ $H_B = -34.5 \text{ kN}$ 8 $V_A = 23.3 \text{ kN}$ $H_A = -20 \text{ kN}$ $V_B = 36.7 \text{ kN}$	

TUTORIAL 2: Degree of Static Indeterminacy

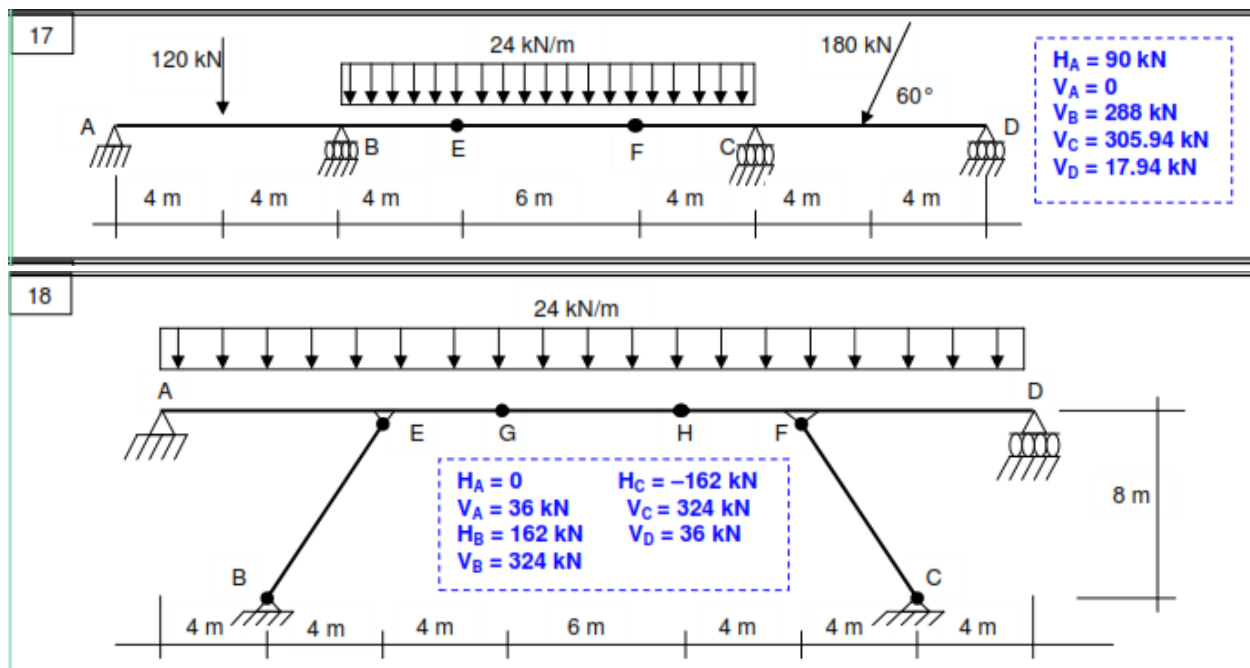
Determine the degree of static indeterminacy of each of the following structures

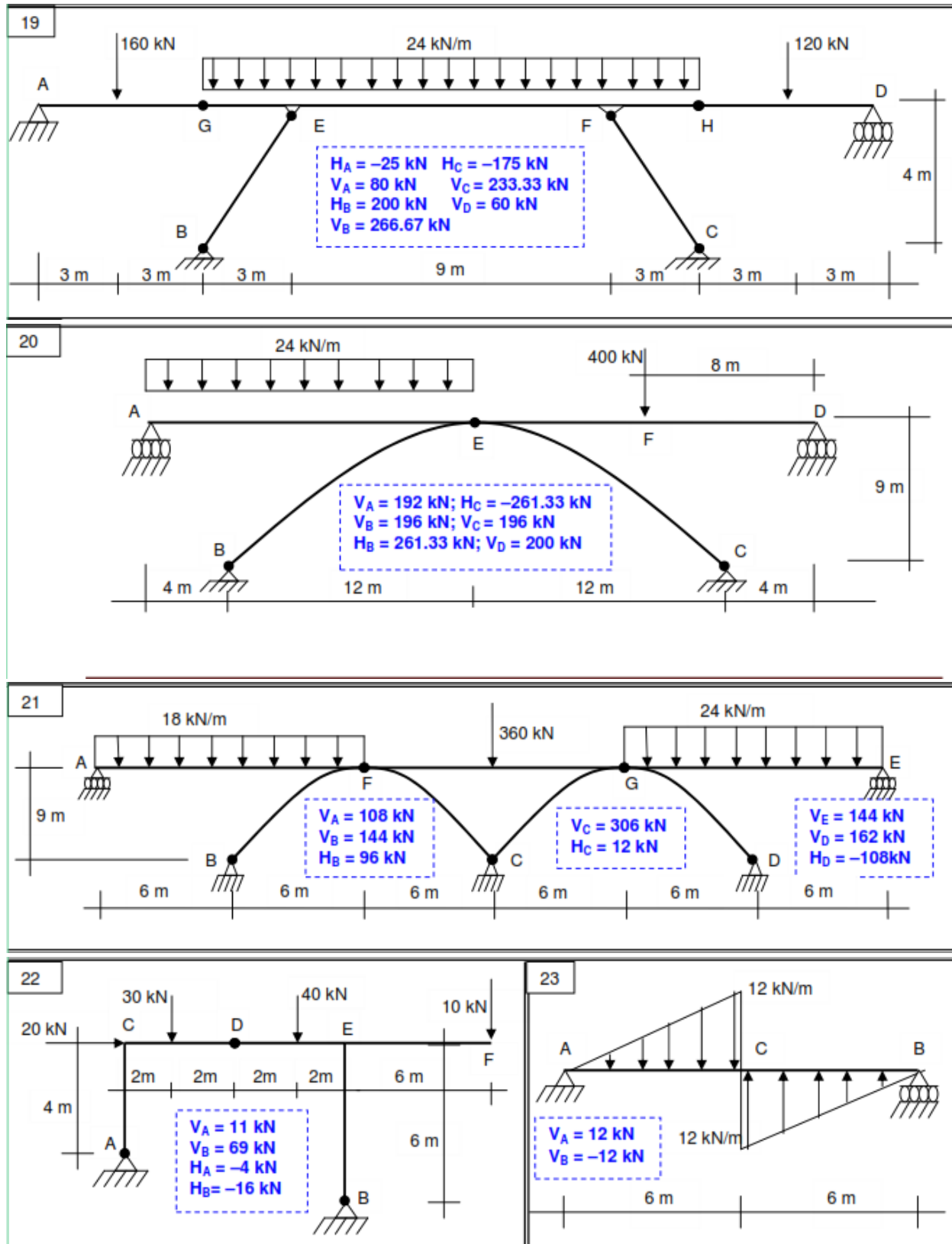
9  $\eta = 0$	10  $\eta = 0$
11  $\eta = 1$	12  $\eta = 12$
13  $\eta = 0$	14  $\eta = 6$



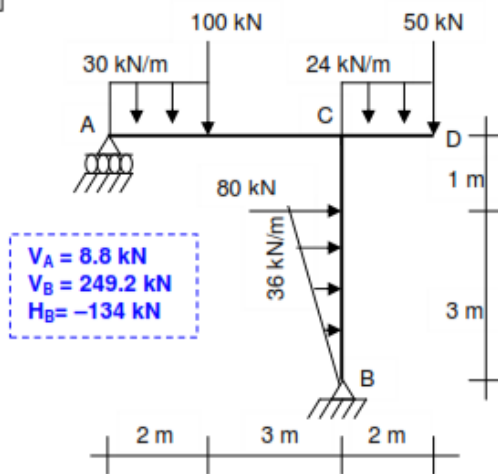
TUTORIAL 3: SUPPORT REACTIONS

Determine the support reactions induced at the supports of each of the structures

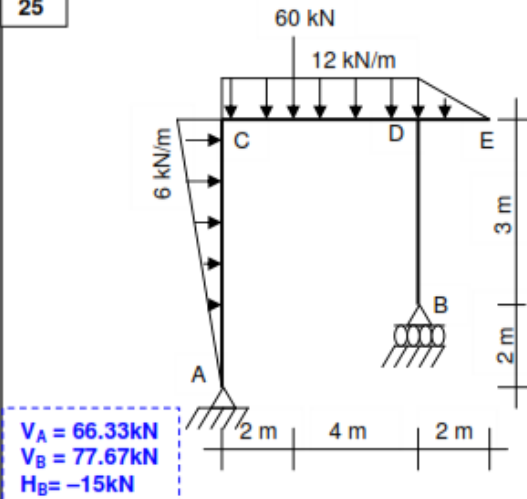




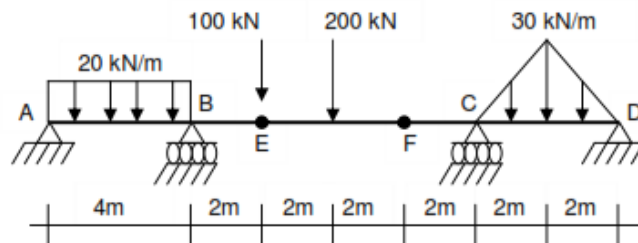
24



25



26



4 CABLES

4.1 Introduction to Cable Structures

A cable transmits the applied loads to the supports by developing axial tension only. The cable is thus referred to as an axial force structure. The material used to make the cable must therefore be of high tensile strength.

Applications of these structures include Hangers (lifts, cranes, maintenance crabs, bridges); Guys (guyed communication masts); stays (cable stayed bridges); Cables (suspension bridges), etc

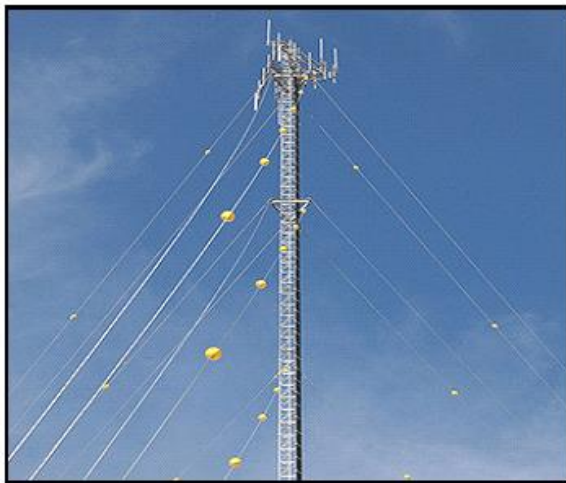


Figure 4-1: Cable structures examples, clockwise from top left, cable crane, cable stayed bridge, guyed communication mast, and suspension bridge

The shape adopted by the cable is a function of the applied loads (ie the shape of the cable will depend on the nature of the loads, their magnitudes, and their points of application). The cable is therefore referred to as a form active structure. It assumes the most efficient geometry for any given

set of loads. Figure 4-2 illustrates some different shapes adopted by the same cable subjected to different loading

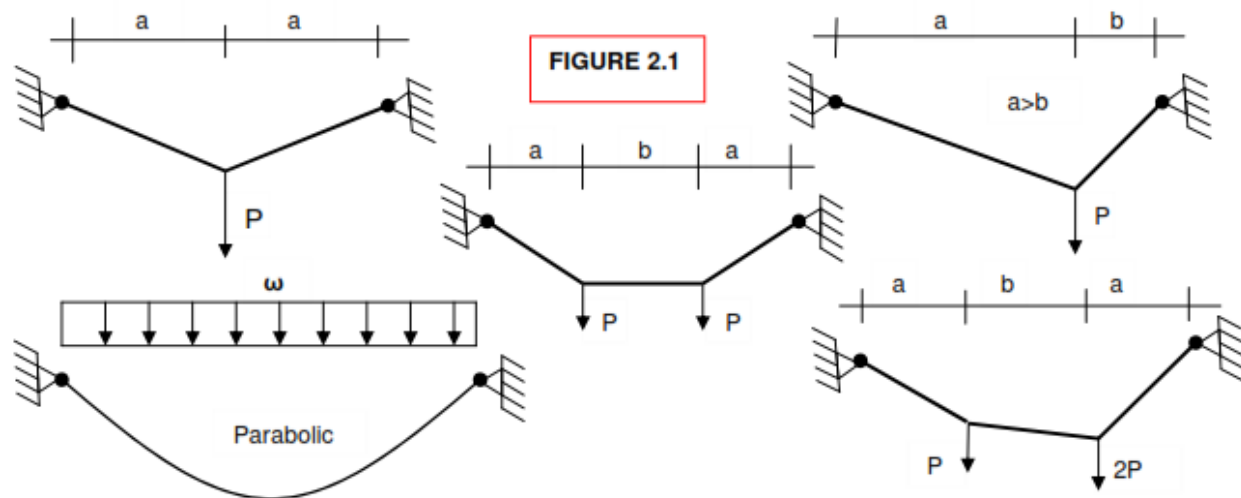


Figure 4-2: Varying shape of cable structure under different loading

The analysis of a cable for any given load scenario involves the determination of the support reactions, the tensile force induced in the cable, the cable profile², the required cable length, and the expected cable elongation³.

The analysis of the cable subjected to point loads relies directly on the principle of static equilibrium, some mathematics, and a bit of common sense. However, the analysis of cables subjected to uniformly distributed loads requires some calculus inputs.

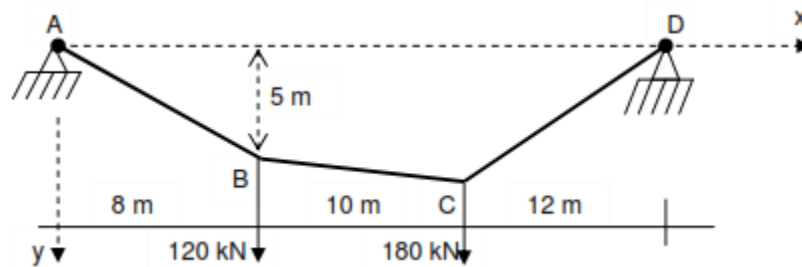
4.2 Cables Subjected to Point Loads

Analyze the cable shown below and determine the following:

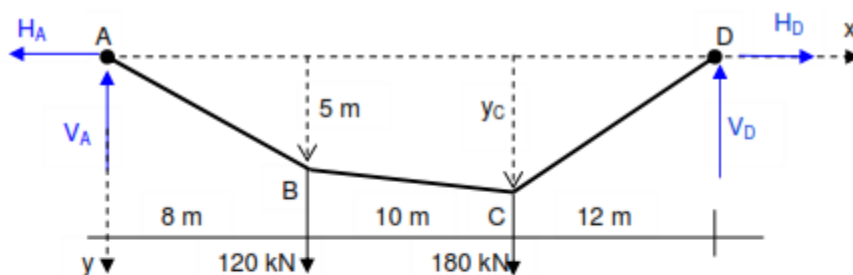
- Support reactions at A and D
- Maximum tension induced in the cable
- The ordinate of the point C
- The required cable length

² This is important especially for structures where the amount of sag may become critical eg transmission lines, cable bridges both suspension and cable stayed

³ This is also important as it directly impacts the amount of total sag experienced by the structure and may have an effect on the load distribution/redistribution on a cable structure

Example 1**Solution****a) Support Reactions**

Isolate the whole cable as a free body diagram



$$\Sigma F_x = -H_A + H_D = 0 \text{ ----- (1)}$$

$$\Sigma F_y = V_A + V_D - 300 = 0 \text{ ----- (2)}$$

$$\Sigma M_A = 120 \times 8 + 180 \times 18 - V_D \times 30 = 0 \text{ ----- (3)}$$

The fourth equation is generated by considering the equilibrium of the portion AB

$$\Sigma M_B (AB) = V_A \times 8 - H_A \times 5 = 0 \text{ ----- (4)}$$

Solving: - $V_A = 160 \text{ kN}$; $H_A = 256 \text{ kN}$; $V_D = 140 \text{ kN}$; $H_D = 256 \text{ kN}$.

b) Maximum Tension

Since H is constant, the maximum tension occurs when the vertical component V attains its highest value. This will be in the portion AB where $V_A = 160 \text{ kN}$

$$\text{Hence } T_{\max} = T_{AB} = \sqrt{(H_A^2 + V_A^2)} = \sqrt{(256^2 + 160^2)} = 301.9 \text{ kN}$$

c) Ordinate of Point C

Isolate the portion C to D,

$$\Sigma M_{C(D)} = (H_D \times y_C) - (V_D \times 12) = 0$$

Therefore;
 $256 y_C = 12(140)$
 Or

$$y_C = 6.563 \text{ m}$$

d) Cable Length

By Pythagoras theorem:

$$L_{AB} = \sqrt{(8^2 + 5^2)} = 9.434 \text{ m}$$

$$L_{BC} = \sqrt{(10^2 + 1.563^2)} = 10.121 \text{ m}$$

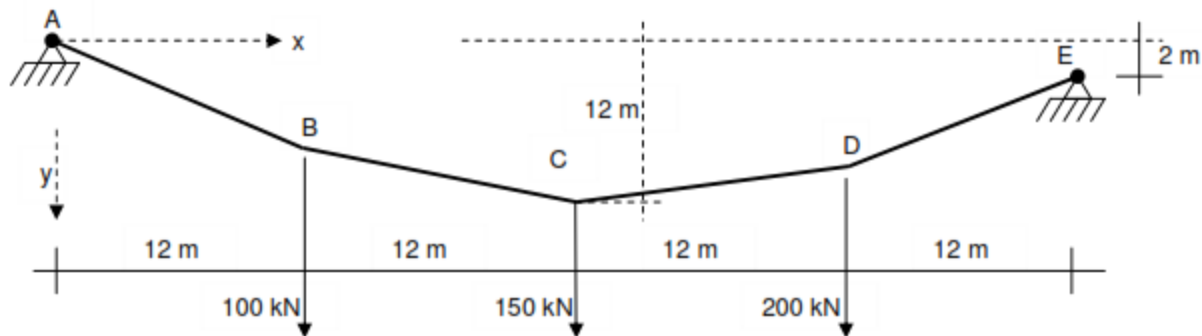
$$L_{CD} = \sqrt{(12^2 + 6.563^2)} = 13.677 \text{ m}$$

$$\text{Hence } L = L_{AB} + L_{BC} + L_{CD} = 33.232 \text{ m}$$

Example 2

Analyse the cable shown below and determine

- The support reactions at A and E
- The maximum tension induced in the cable
- The ordinate of points B and D
- The required cable length



Solution

- (a) Isolate the whole cable; then isolate the portion C – D – E

$$V_A = 213.64 \text{ kN}; H_A = 327.27 \text{ kN}; V_E = 236.36 \text{ kN}; \text{ and } H_E = 327.27 \text{ kN.}$$

$$(b) T_{\max} = T_{DE} = \sqrt{(H_E^2 + V_E^2)} = \sqrt{(327.27^2 + 236.36^2)} = 403.70 \text{ kN}$$

$$(c) y_B = 7.834 \text{ m}; y_C = 12 \text{ m}; \text{ and } y_D = 10.667 \text{ m [All measured from level of A]}$$

$$(d) L_{AB} = 14.331 \text{ m}; L_{BC} = 12.703 \text{ m}; L_{CD} = 12.074 \text{ m}; \text{ and } L_{DE} = 14.803 \text{ m}$$

$$\text{Hence } L = 53.918 \text{ m}$$

4.3 Cable Subjected to Uniformly Distributed Load

Figure 4-3 below shows the general model, dimensions and notations, for cable subjected to a uniformly distributed load ω .

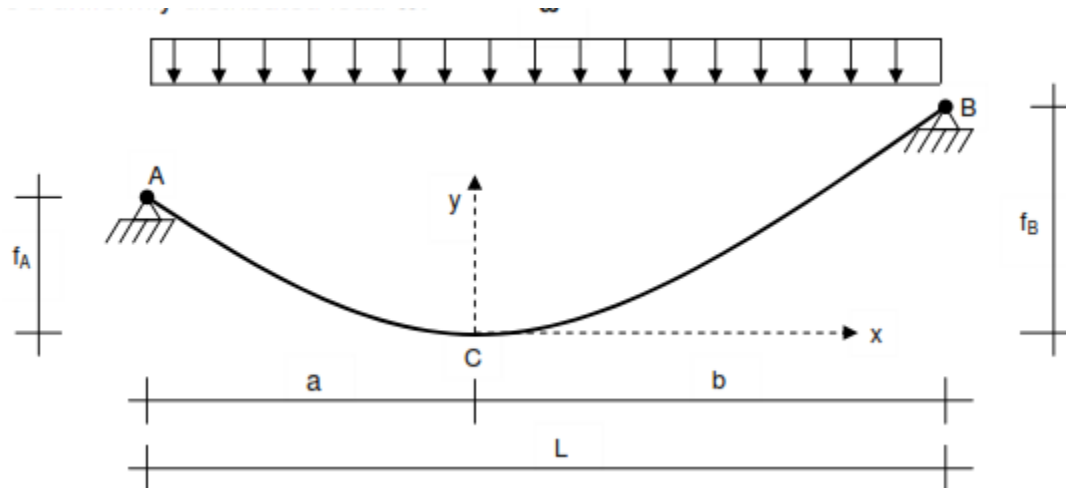


Figure 4-3: Cable under uniformly distributed load

The reference x-y coordinate system is chosen to have its origin at the lowest point C so that the cable profile conforms to a **perfect parabola**.

4.3.1 Cable Profile

Consider a free body diagram of an element (ie a small portion) of the cable (which is magnified for clarity only) as shown in Figure 4-4 below.

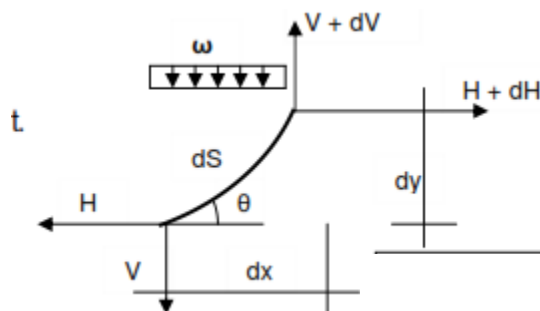


Figure 4-4: Cable free body diagram

The following assumptions are made:

1. The element is small enough to be considered straight
2. The cable self weight is negligible

For this free body to be in static equilibrium:

- $\Sigma F_x = -H + H + dH = 0$
Thus $dH = 0$, which means that H remains constant throughout the length of the cable
- $\Sigma F_y = -V + V + dV - \omega dx = 0$, Thus $dV = \omega dx$
 $\omega = dV/dx$ (i)
- In addition to the principle of compatibility requires that $\tan \theta = V/H = dy/dx$,
Hence we can express;
 $V = H(dy/dx)$ (ii)
- Combining equations (i) and (ii), we obtain $\omega = d/dx \{H dy/dx\}$
Hence
 $\omega/H = d^2y/dx^2$ (iii)

Notice that in equation (iii) both ω and H are constants and the equation (iii) is therefore a standard 2nd order differential equation whose solution may be obtained by integrating twice and applying the existing boundary conditions

Thus by integrating equation (iii) twice we obtain:

- $dy/dx = (\omega/H)x + C_1$, and
- $y = (\omega/2H)x^2 + C_1x + C_2$

Referring to the x-y coordinate system shown in Figure 4-3 it can be seen that the existing boundary conditions are:

- $y = 0$ when $x = 0$, so that $C_2 = 0$, and
- $dy/dx = 0$ when $x = 0$, so that $C_1 = 0$

Hence the cable profile is given by the equation

$$y = \omega x^2 / 2H \quad \text{(iv)}^4$$

Equation (iv) is the equation of a perfect parabola: $y = kx^2$; where $k = (\omega/2H)$

4.3.2 Maximum and Minimum Tension

Consider the free body diagram shown in Figure 4-4, it was noted in eqn (i) that

$\omega = dV/dx$ which can re-arranged as following:

⁴ This equation is used to evaluate a , b , and H

$$dV = \omega dx \quad (v)$$

The equation (v) can be solved by integrating once, and applying the existing boundary conditions, thus;

$$V = \omega x + C_3 \quad (vi)$$

Referring to the cable profile shown in Figure 4-3, it can be seen that at the point C the cable is horizontal and hence the tensile force in the cable is due to the horizontal component H only. Thus the existing boundary condition is that:

$$V = 0 \text{ when } x = 0, \text{ so that } C_3 = 0$$

Hence we express:

$$V = \omega x \quad (vii)^5$$

Equation (vii) shows that V varies linearly with the distance x as measured from point C, and is used to evaluate V_A and V_B .

The tension at any part on the cable is given by:

$$T = \sqrt{(H^2 + V^2)}$$

Since H is constant, the tension at any point on the cable can be expressed as;

$$T = \sqrt{(H^2 + (\omega x)^2)} \quad (viii)$$

Equation (viii) shows that maximum tension occurs when x attains its largest value

4.3.3 Support Reactions

The horizontal component $H = H_A$ and H_B is obtained by applying **Equation (iv)**⁶ to the given control points. The vertical reaction components V_A and V_B are obtained from **Equation (vii)**⁷. This approach is based on the second principle.

⁵ This equation is used to evaluate V_A and V_B

⁶ $y = \omega x^2 / 2H$

⁷ $V = \omega x$

Alternatively, once a and b are known the reaction components can be obtained by applying the equations of equilibrium on a free body diagram of the whole cable. This approach is based on first principles

4.3.4 Cable Length: Exact Formulae

Consider once again the free body diagram shown in Figure 4-4. The elemental length is given as ds which may be expressed as:

$$ds = \sqrt{\{(dx)^2 + (dy)^2\}} = \sqrt{1 + (dy/dx)^2} dx$$

The cable length is given by the integral of ds over the whole span thus:

$$L = \int_L ds = \int_L \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx \quad \text{(ix)}$$

Since $y = (\omega/2H)x^2$, then $dy/dx = (\omega x/H)$, so that

$$ds = \sqrt{1 + (\omega x/H)^2} dx \quad \text{(x)}$$

The cable length from point C to point B is calculated using integral calculus as

$$L_{CB} = \int_0^b \sqrt{1 + \left(\frac{\omega x}{H}\right)^2} dx \quad \text{(xi)}$$

Equation (xi) is a standard equation in calculus, whose solution is given as

$$L_{CB} = \frac{1}{2} b \sqrt{1 + \left(\frac{\omega b}{H}\right)^2} + \left(\frac{H}{2\omega}\right) \sinh^{-1}\left(\frac{\omega b}{H}\right) \quad \text{(xii)}$$

Similarly;

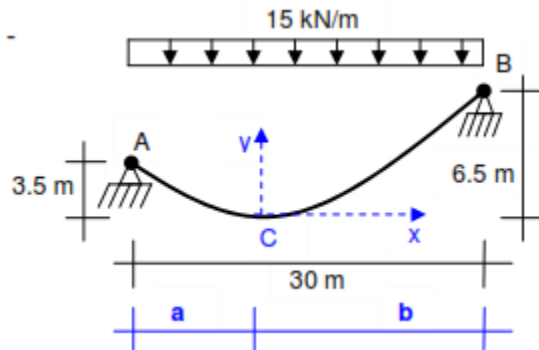
$$L_{CA} = \frac{1}{2} a \sqrt{1 + \left(\frac{\omega a}{H}\right)^2} + \left(\frac{H}{2\omega}\right) \sinh^{-1}\left(\frac{\omega a}{H}\right) \quad \text{(xii)}^8$$

⁸ This equation is referred to as the exact formula and is used to calculate the required cable length for any cable

Example 3

Consider the cable shown below, determine:

- The support reactions
- The maximum and minimum tension induced in the cable
- The required cable length

**Solution****(a) Support Reactions**

Let A to C be “a”, and C to B be “b”

Given $f_A = 3.5\text{m}$, $f_B = 6.5\text{m}$, $\omega = 15\text{ kN/m}$, and $L = 30\text{m}$

Using equation (iv);

$$y = \omega x^2 / 2H$$

We have, $f_A = \omega a^2 / (2H)$ and $f_B = \omega b^2 / (2H)$ which may be rearranged to make H the subject, thus:

$$H = \omega a^2 / (2f_A), \text{ and } H = \omega b^2 / (2f_B)$$

Dividing one equation by the other gives:

$$(a/b)^2 = f_A / f_B \quad (a)$$

Compatibility requires that

$$a + b = 30 \quad (b)$$

Solving equations (a) and (b) simultaneously gives

$$a = 12.697\text{m}, b = 17.303\text{m} \text{ and } H = 345.46\text{ kN}$$

The vertical support reaction components are obtained using equation (vii), thus

$$V_A = \omega a = 190.46\text{ kN}, \text{ and } V_B = \omega b = 259.55\text{ kN}$$

(b) The maximum tension

Maximum tension occurs at B and is given by

$$T_{\max} = T_B = \sqrt{\{H^2 + V_B^2\}} = 432.10 \text{ kN}$$

The minimum tension is given by

$$T_{\min} = H = 345.46 \text{ kN}$$

(c) Cable Length

$$L_{CA} = (a/2)\sqrt{1+(\omega a/H)^2} + (H/2\omega) \sinh^{-1}(\omega a/H) = 13.314 \text{ m}$$

$$L_{CB} = (b/2)\sqrt{1+(\omega b/H)^2} + (H/2\omega) \sinh^{-1}(\omega b/H) = 18.815 \text{ m}$$

Hence

$$L_{ACB} = 13.314 + 18.815 = 32.129 \text{ m}$$

4.3.5 Cable Length

Consider the case of a symmetrical cable as shown in Figure 4-5

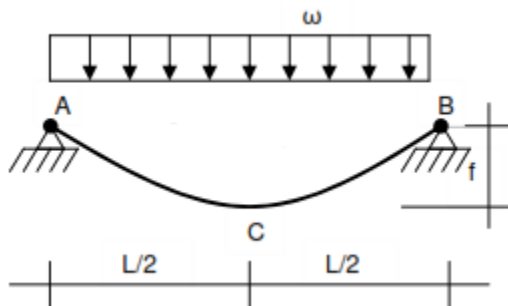


Figure 4-5: Symmetrical cable

In this case $a = b = L/2$, and $f_A = f_B = f$, say

Using equation (iv) $y = \omega x^2/2H = (\omega/2H)x^2$, we can write

$$f = (\omega/2H)(L/2)^2 = \omega L^2/8H \quad \text{(xiii)}$$

Rearranging equation (xiii) we obtain

$$H = \omega L^2 / 8f \quad (\text{xiv})$$

Using equation (viii), $T = \sqrt{(H^2 + (\omega x)^2)}$, we can write;

$$T_A = T_B = T_{\max} = \sqrt{(H^2 + (\omega L/2)^2)} \quad (\text{xv})$$

By substituting (xiv) into (xv) and simplifying we obtain

$$T_{\max} = T_A = T_B = (\omega L^2 / 8f) \sqrt{1 + 16(f/L)^2} \quad (\text{xvi})$$

Using equation (xiv) we can write

$$\omega / 2H = 4fL^2 \quad (\text{xvii})$$

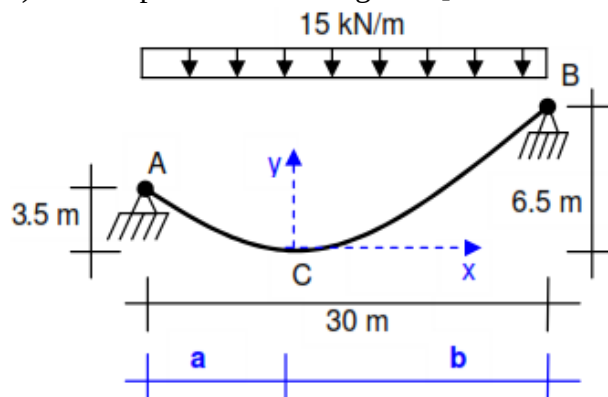
Following this thought process, it can be shown that if $f/L \ll 1$, and the cable length can be approximated by:

$$L_{CA} = L_{CB} = (L/2) \{1 + (8/3)(f/L)^2\} \quad (\text{xviii})^9$$

Example 3

Consider the cable shown below, determine:

- The support reactions
- The maximum and minimum tension induced in the cable
- The required cable length, using both the exact and approximate formulae
- The expected cable elongation [Take $E = 200 \text{ kN/mm}^2$ and $A = 576 \text{ mm}^2$]



⁹ Equation (xviii) is referred to as the approximate formulae for cable length. Equations (xviii) and (xii) can be used to calculate the cable length. It is however more ideal to work with equation (xii)

Solution**(a) Support reactions**

Let A to C be “a” and C to B be “b”

Given $f_A = 3.5\text{m}$, $f_B = 6.5\text{m}$, $\omega = 15 \text{ kN/m}$, and $L = 30\text{m}$;

Using equation (iv),

$y = \omega x^2 / 2H$, we have;

$$f_A = \omega a^2 / (2H) \text{ and } f_B = \omega b^2 / (2H)$$

Which may be rearranged to make H the subject, thus;

$$H = \omega a^2 / (2f_A) \text{ and } H = \omega b^2 / (2f_B)$$

Dividing one equation with the other give;

$$(a/b)^2 = f_A / f_B \quad (1)$$

Compatibility requires that:

$$a + b = 30 \quad (2)$$

Solving equations (1) and (2) simultaneously gives;

$$a = 12.697 \text{ m}, b = 17.303 \text{ m}, \text{ and } H = 345.46 \text{ kN} = H_A = H_B$$

The vertical support reactions are obtained using equation (viii), thus;

$$V_A = \omega a = 190.46 \text{ kN}; \text{ and } V_B = \omega b = 259.55 \text{ kN}$$

(b) Maximum tension

Maximum tension occurs at B and is given by

$$T_{\max} = T_B = \sqrt{H^2 + V_B^2} = 432.10 \text{ kN}$$

Minimum tension is given by;

$$T_{\min} = H = 345.46 \text{ kN}$$

(c) Cable Length Using Exact Formula

$$L_{CA} = (a/2) \sqrt{1 + (\omega a/H)^2} + (H/2\omega) \sinh^{-1} (\omega a/H) = 13.314 \text{ m}$$

$$L_{CB} = (b/2) \sqrt{1 + (\omega b/H)^2} + (H/2\omega) \sinh^{-1} (\omega b/H) = 18.815 \text{ m}$$

$$\text{Hence } L_{ACB} = 13.314 + 18.815 = 32.129 \text{ m}$$

(d) Cable Length Using Approximate Formulas

$$L_{CA} \approx a \{1 + (8/3) (f_A/L)^2\} \approx 13.158 \text{ m and } L_{CB} \approx b \{1 + (8/3) (f_B/L)^2\} \approx 19.469 \text{ m}$$

$$\text{Hence } L_{ACB} \approx 13.158 + 19.469 \approx 32.627 \text{ m}$$

The percentage error, $e = 1.55\%$ on the safe side

4.4 Cable Elongation

4.4.1 Basic Principle

When a cable is subjected to loading, the material develops stresses which result in strains which make the cable longer than theoretically predicted in the preceding sections. Such cable elongation may be predicted as described below:

The underlying principle is based on the uni-axial tensile test model illustrated in **Figure *****

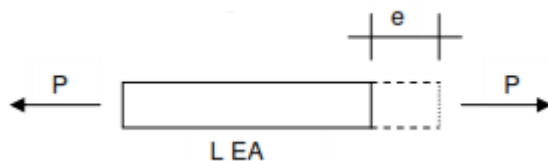


Figure *:** Cable elongation

The stress is given by $\sigma = P/A$

The strain is given by $\epsilon = e/L$

By the principle of constitutivity, $\sigma = E\epsilon$

So that $P/A = E(e/L)$ and hence,

$$e = PL/EA$$

The quantity EA is called the axial stiffness

4.4.2 Cable Subjected to Point Loads

Each straight point of the cable will suffer an extension is given by $e = TL/EA$

The elongation of the cable subjected to point loads is therefore calculated by piece-wise summation, thus;

$$\Delta s = \sum_{i=1}^n T_i L_i / EA \quad (\text{xix})$$

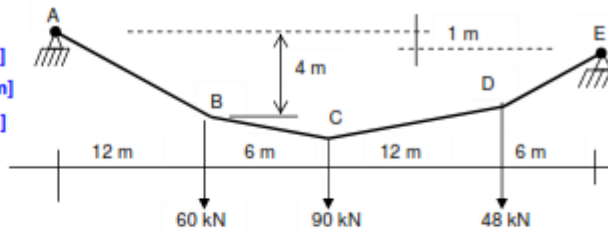
4.4.3 Cable Subjected to UDL

TUTORIAL 4: CABLES

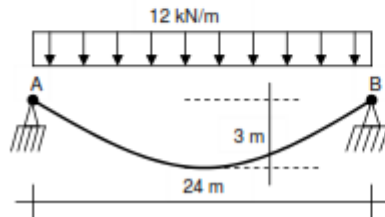
1

Determine: -

- (a) T_{max} [320.8 kN, at A]
 (b) y_C and y_D [4.817 m, 2.903 m]
 (c) Req'd cable length [37.151 m]
 (d) Cable elongation [160.5 mm]

Take $EA = 72.74 \times 10^3 \text{ kN}$ 

2



Determine: -

- (a) T_{max} [322 kN]
 (b) Cable length [Exact = 24.965 m
Approx = 25.000 m]
 (c) Cable elongation [62 mm]

Take $EA = 120 \times 10^3 \text{ kN}$

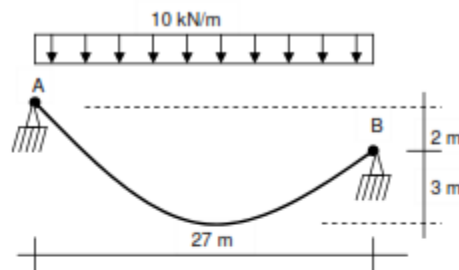
3

Determine: -

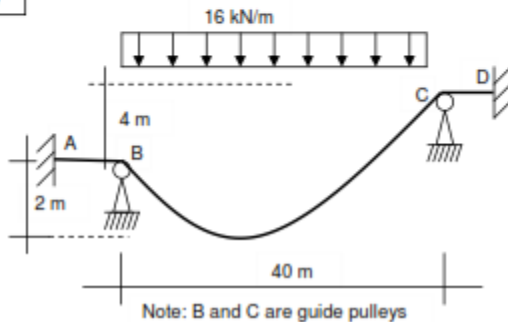
- (a) T_{max} [277.02 kN]
 (b) Cable length [Exact = 28.524 m
Approx = 28.605 m]
 (c) Cable elongation [58 mm]

Take $EA = 120 \times 10^3 \text{ kN}$

[a = 15.215 m, b = 11.785 m]



4



Determine: -

- (a) T_{max} , T_{min} [948.5 kN, 857.4 kN]
 (b) Support reaction components at B and C [$V_B = 234.2 \text{ kN}$
 $V_C = 405.7 \text{ kN}$
 $H_B = 31.4 \text{ kN}$
 $H_C = 91.1 \text{ kN}$]

5 Determine the weight W required to make the cable sag at C be 1 m as shown. [$W = 2276.84 \text{ kN}$]

Note: Support at B is a guide pulley

6

Determine: -

(a) T_{max} in DEB [75.12 kN]

(b) The sag f [7 mm]

Note: DEB is a guy supporting the frame ABC

7

Determine: -

(a) T_{max} [2447.5 kN]

(b) Sag f [2.25 m]

(c) Cable length [$\text{Exact} = 60.298 \text{ m}$, $\text{Approx} \approx 60.300 \text{ m}$]

[$a = 20 \text{ m}$, and $b = 40 \text{ m}$]

8 If $T_{max} \leq 500 \text{ kN}$ determine the approximate required cable length [25.684 m]

9 Determine: -

(a) The sag f_c of the cable span CDE [625 mm]

(b) T_{max} , stating where it occurs [263.88 kN] [Just to the left of C]

(c) Cable length [$\text{Exact} = 25.353 \text{ m}$] [Approx $\approx 25.364 \text{ m}$]

Support at C is a pulley on rollers

5 THE ARCH

5.1 Introduction

The shape of an ideal arch is the inverse of the shape of a cable subjected to the same loads. Such an ideal arch would be a purely compression structure. The arch is not form-active. In practice, the ideal arch profile is hard to achieve because of the required span to height ratio. Most arches are therefore non ideal, and will carry the applied loads by developing not only axial compression, but also shear force, and bending moment.



Figure 5-1: Arch structures

The masonry arch is amongst the earliest forms of construction. Today arches may be constructed using masonry, reinforced concrete, steel, laminated timber or any combination thereof.

Arches find applications in many areas including industrial low rise sheds, highway bridges, foot bridges, large span roof systems, architectural works in churches, mosques, libraries, civil block, etc. Arches may make various geometric shapes as illustrated in Figure 5-2 (a). The parabolic shape is very efficient for uniformly distributed loading. The parabolic arch transmits the uniformly distributed loading by developing axial stresses only; no shear, no moments.

Arches may be constructed with varying degrees of static indeterminacy, as illustrated in Figure 5-2 (b) below. In this course, only statically determinate problems will be discussed.

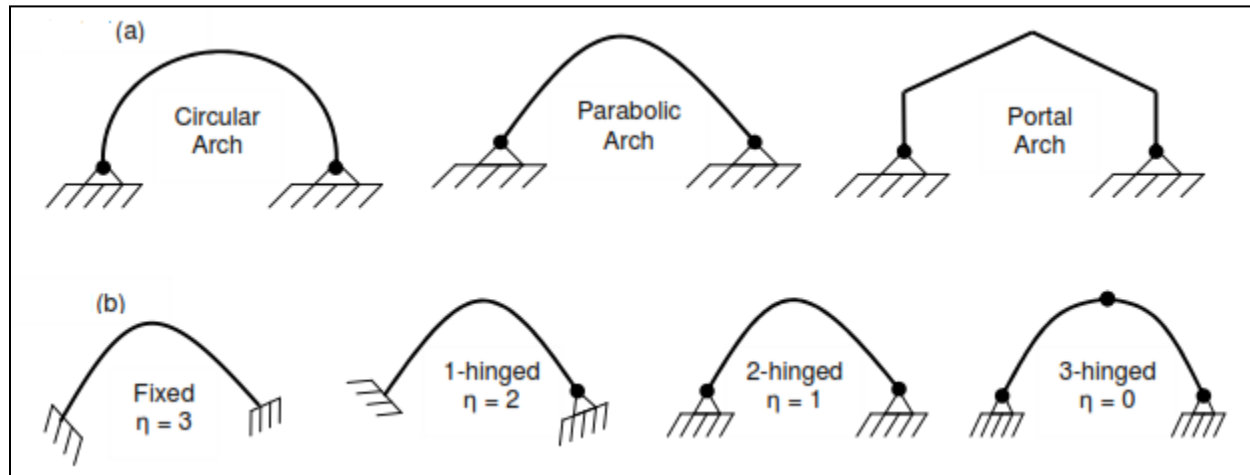


Figure 5-2: Arches – Different degrees of static indeterminacy

5.2 Structural Analysis

Structural analysis of arches involves determining;

1. The arch profile
2. The support reactions
3. The axial force N , shear force V , and the bending moment M induced at any given cross-section
4. The displacement of the salient points on the arch profile

Once the arch profile is determined, items 2 and 3 of the analysis can be achieved by application of the principle of static equilibrium

The general procedure for the analysis of arches may be outlined as below;

1. Set up an appropriate x - y coordinate system and determine the arch profile. Hence determine, at any given cross section, the coordinates and the slope.
2. Evaluate the support reactions
3. For any given cross section;
 - i. Isolate an appropriate free body diagram
 - ii. Apply equilibrium to generate three equations in N , V , and M
 - iii. Solve simultaneously

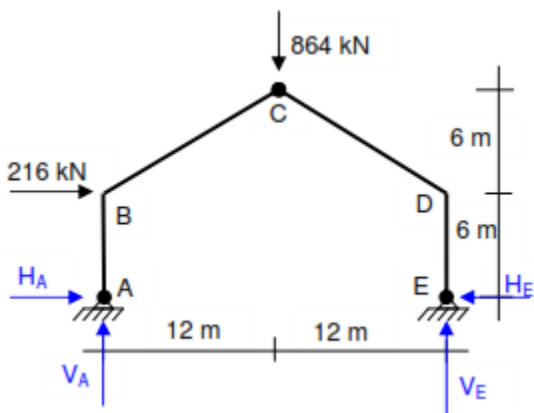
The above procedure is illustrated in the examples 1, 2, and 3 below

The axial force N , shear force V , and bending moment M are continuous (at least piece-wise) mathematical functions over the span of the arch. These functions may be shown as graphical variations of the load effect under consideration, from cross section along the arch or along the span. Such graphical representations are referred to as diagrams. The axial force diagram **AFD**, shear force diagram **SFD**, bending moment diagram **BMD**, and the deformed shape, **D-Shape**, are dealt with in later chapters of this discussion.

Arches are predominantly compression structures, In order to avoid buckling the member sizes are usually bulky. Consequently the arch suffers very small axial deformations. However, in certain arch types, the displacements due to flexure may be significant.

Example 1

Determine N, V, and M induced at points B and D on the pitched portal arch shown below¹⁰



Solution

Support reactions

$$\Sigma F_x = H_A - H_E + 216 = 0$$

$$\Sigma F_y = V_A + V_E - 864 = 0$$

$$\Sigma M_A = 216(6) + 864(12) - V_E(24) = 0$$

$$\Sigma M_{C(R)} = H_E(12) - V_E(12) = 0$$

$$\text{Solving: } V_A = 378 \text{ kN} \quad V_E = 486 \text{ kN}$$

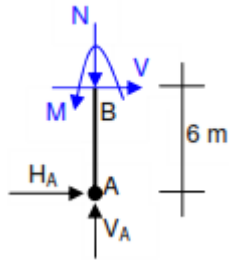
$$H_A = 270 \text{ kN} \quad H_E = 486 \text{ kN}$$

NB

- The values of N and V will change as you arrive and pass the point B (or point D). We will therefore think of just below and just above the points B and D, where the word “just” assumes zero distance from the point of interest.
- It's also important to note that the solution of this problem from this point will consider resolution of forces in the directions of the unknowns; N and V.

Just below joint B;

¹⁰ It's important to note that in this example a pin has been introduced at point C to make the arch statically determinate. Without the pin, the arch would be statically indeterminate (with four unknown reactions) and only three equations available for solution.

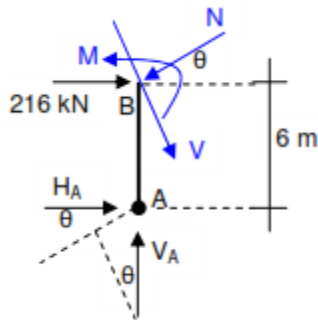


$$\Sigma F_N = N - V_A = 0 \Rightarrow N = V_A = 378 \text{ kN}$$

$$\Sigma F_V = V + H_A = 0 \Rightarrow V = -H_A = -270 \text{ kN}$$

$$\Sigma M_X = H_A(6) + M = 0 \Rightarrow M = -H_A(6) = -1620 \text{ kNm}$$

Just above point B¹¹ ($\theta = \tan^{-1}(6/12) = 26.57^\circ$)



$$\Sigma F_N = N - V_A \sin \theta - H_A \cos \theta - 216 \cos \theta = 0$$

$$\Rightarrow N = V_A \sin \theta + H_A \cos \theta + 216 \cos \theta = 603.7 \text{ kN}$$

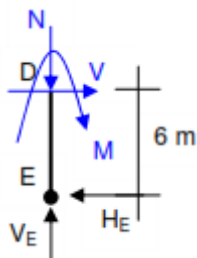
$$\Sigma F_V = V - V_A \cos \theta + H_A \sin \theta + 216 \sin \theta = 0$$

$$\Rightarrow V = V_A \cos \theta - H_A \sin \theta - 216 \sin \theta = 120.7 \text{ kN}$$

$$\Sigma M_X = H_A(6) + M = 0$$

$$\Rightarrow M = -H_A(6) = -1620 \text{ kNm}$$

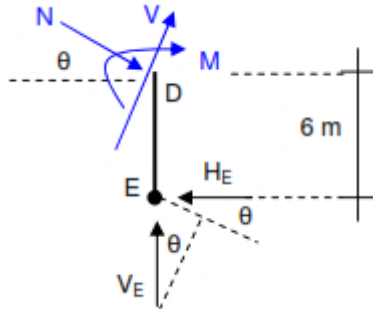
Just below joint D



¹¹ Important to note here that the moment “just above” and “just below” the point B is of equal magnitude. This is an important frame continuity requirement, for the case where no moment is applied at the joint B. The same case applies to joint D as well.

$$\begin{aligned}\Sigma F_N &= N - V_E = 0 \quad \Rightarrow \quad N = V_E = 486 \text{ kN} \\ \Sigma F_V &= V - H_E = 0 \quad \Rightarrow \quad V = H_E = 486 \text{ kN} \\ \Sigma M_X &= H_E(6) + M = 0 \quad \Rightarrow \quad M = -H_E(6) = -2916 \text{ kNm}\end{aligned}$$

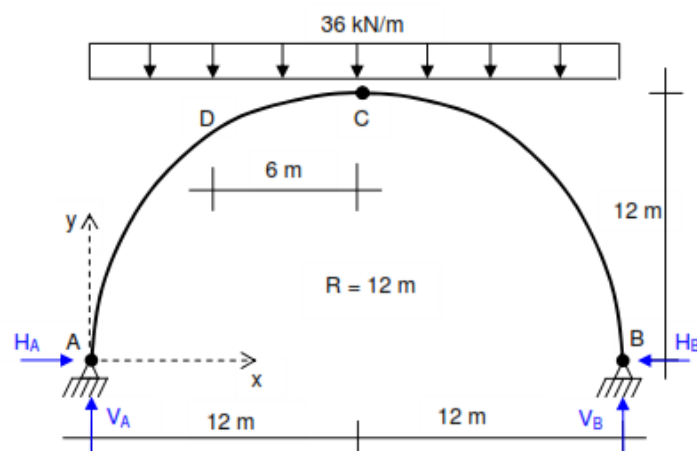
Just above Joint D ($\theta = \tan^{-1} (6/12) = 26.57^\circ$)



$$\begin{aligned}\Sigma F_N &= N - V_E \sin \theta - H_E \cos \theta = 0 \\ \Rightarrow N &= V_E \sin \theta + H_E \cos \theta = 652.1 \text{ kN} \\ \Sigma F_V &= V + V_E \cos \theta - H_E \sin \theta = 0 \\ \Rightarrow V &= -V_E \cos \theta + H_E \sin \theta = -217.3 \text{ kN} \\ \Sigma M_X &= H_E(6) + M = 0 \\ \Rightarrow M &= -H_E(6) = -2916 \text{ kNm}\end{aligned}$$

Example 2

Determine the N, V and M at point D in the semi-circular arch shown below¹²



¹² Note that a pin has been introduced at C to make this problem statically determinate. Without the pin, this problem would be statically indeterminate

Solution

Support reactions

$$\Sigma F_x = H_A - H_B = 0$$

$$\Sigma F_y = V_A + V_B - 36(24) = 0$$

$$\Sigma M_A = 36(24)(12) - V_B(12) = 0$$

$$\Sigma M_{C(R)} = (36)(12)(6) + H_B(12) - V_B(12) = 0$$

Solving;

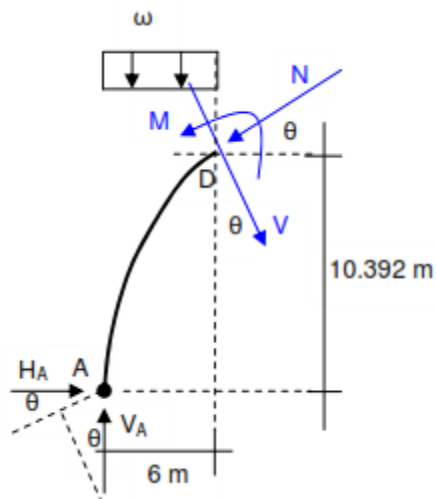
$$V_A = 432 \text{ kN}, H_A = 216 \text{ kN}, V_B = 432 \text{ kN}, H_B = 216 \text{ kN}$$

Arch profileUsing the x-y system shown on the figure¹³;

$$y = \sqrt{(2Rx - x^2)}$$

$$dy/dx = (R - x)/\sqrt{(2Rx - x^2)}$$

$$\text{Hence, } \theta = \tan^{-1}\{(R - x)/\sqrt{(2Rx - x^2)}\}$$

N, V, and M at D

$$y = \sqrt{(2Rx - x^2)} = 10.392\text{m}$$

¹³ These equations are derived from the basic equation for a circle; $r^2 = (x - h)^2 + (y - k)^2$, where (h,k) are the coordinates of the centre of the circle

$$\theta = \tan^{-1}\{(R - x)/\sqrt{(2Rx - x^2)}\} = 30^\circ$$

$$\Sigma F_N = N - V_A \sin \theta - H_A \cos \theta + 36(6) \sin \theta = 0$$

$$\Rightarrow N = V_A \sin \theta + H_A \cos \theta - 36(6) \sin \theta = 295.1 \text{ kN}$$

$$\Sigma F_V = V - V_A \cos \theta + H_A \sin \theta + 36(6) \cos \theta = 0$$

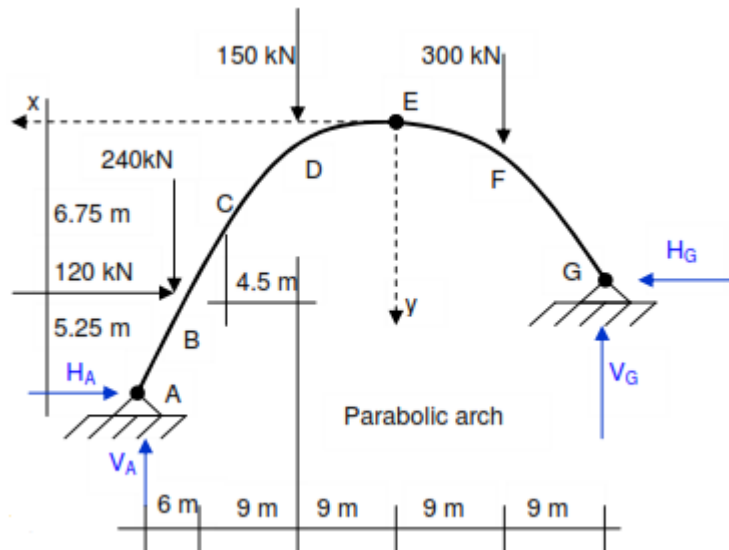
$$\Rightarrow V = V_A \cos \theta - H_A \sin \theta - 36(6) \cos \theta = 79.1 \text{ kN}$$

$$\Sigma M_X = V_A(6) - H_A(10.392) - 36(6)(3) - M = 0$$

$$\Rightarrow M = V_A(6) - H_A(10.392) - 36(6)(3) = -300.7 \text{ kN}$$

Example 3

Determine N, V, and M induced at A, B, C, D, E, F, and G in the parabolic arch shown below



Solution

Using the x-y coordinate system shown, the arch profile is given by

$$y = kx^2$$

At A, y is 12m, and x 24m

Thus;

$$12 = k(24)^2; k = 1/48$$

Hence $y = x^2/48$, and $dy/dx = x/24$

At D, $x = 9\text{m}$; therefore $y_D = 1.6875\text{m}$

Also at F, $x = 9\text{m}$, therefore $y_F = 1.6875\text{m}$

At C, $x = 13.5\text{m}$, therefore $y_C = 3.7969\text{m}$

At B, $x = 18\text{m}$, therefore $y_B = 6.75\text{m}$, same as at G

Isolate the whole arch as a free body diagram

$$\sum F_x = H_A + 120 - H_G = 0$$

$$\sum F_y = V_A + V_G - 690 = 0$$

$$\sum M_A = 120 \times 5.25 + 240 \times 6 + 150 \times 15 + 300 \times 33 - V_G \times 42 - H_G \times 5.25 = 0$$

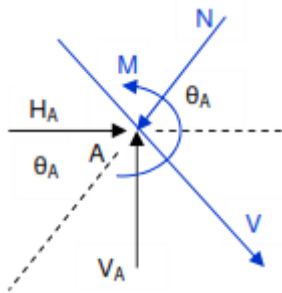
$$\sum M_{C(R)} = 300 \times 9 + H_G \times 6.75 - V_G \times 18 = 0$$

Solving: **$V_A = 398.6 \text{ kN}$, $H_A = 257.1 \text{ kN}$**

$V_G = 291.4 \text{ kN}$ and $H_G = 377.1 \text{ kN}$

At A

Isolate a fbd of the arch portion just above point A



$$\tan \theta_A = dy/dx \big|_A = 24/24 \Rightarrow \theta_A = 45^\circ$$

$$\sum F_N = N - V_A \sin \theta_A - H_A \cos \theta_A = 0$$

$$\Rightarrow \mathbf{N = 463.6 \text{ kN}}$$

$$\sum F_V = V - V_A \cos \theta_A + H_A \sin \theta_A = 0$$

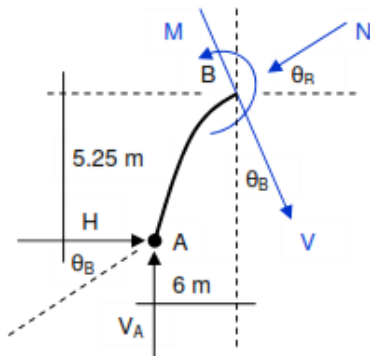
$$\Rightarrow \mathbf{V = 100.1 \text{ kN}}$$

$$\sum M_A = \mathbf{M = 0}$$

Just to the left of B¹⁴

Isolate a fbd of the arch portion A to B (the loads at B will not feature)

¹⁴ Note that the moments just to the left of B and just to the right of B are equal. This is necessary for the continuity of the frame



$$\tan \theta_B = dy/dx \big|_B = 18/24 \Rightarrow \theta_B = 36.87^\circ$$

$$\sum F_N = N - V_A \sin \theta_B - H_A \cos \theta_B = 0$$

$$\Rightarrow \mathbf{N = 444.4 \text{ kN}}$$

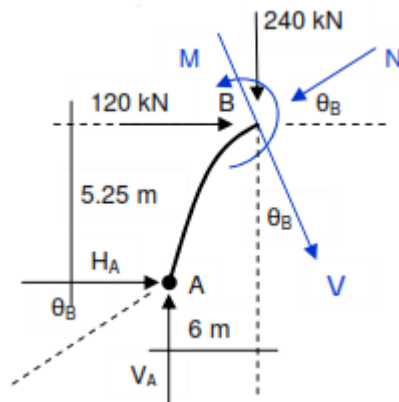
$$\sum F_V = V - V_A \cos \theta_B + H_A \sin \theta_B = 0$$

$$\Rightarrow \mathbf{V = 164.6 \text{ kN}}$$

$$\sum M_B = V_A \times 6 - H_A \times 5.25 - M = 0$$

$$\Rightarrow \mathbf{M = 1041.8 \text{ kNm}}$$

Just to the right of B



Isolate a fbd of the arch portion A to B (the loads at B will feature)

$$\tan \theta_B = dy/dx \big|_B = 18/24 \Rightarrow \theta_B = 36.87^\circ$$

$$\begin{aligned} \sum F_N = N - V_A \sin \theta_B - H_A \cos \theta_B - 120 \cos \theta_B \\ + 240 \sin \theta_B = 0 \end{aligned} \Rightarrow \mathbf{N = 396.8 \text{ kN}}$$

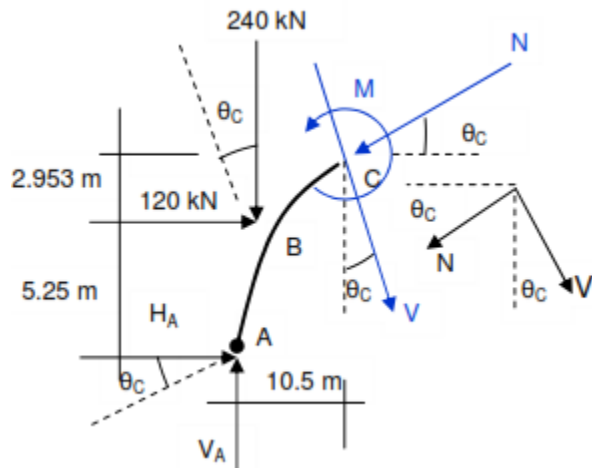
$$\begin{aligned} \sum F_V = V - V_A \cos \theta_B + H_A \sin \theta_B + 120 \sin \theta_B \\ + 240 \cos \theta_B = 0 \end{aligned} \Rightarrow \mathbf{V = -99.4 \text{ kN}}$$

$$\sum M_D = V_A \times 6 - H_A \times 5.25 - M = 0$$

$$\Rightarrow \mathbf{M = 1041.8 \text{ kNm}}$$

At C

Isolate a fbd of the arch portion A to C



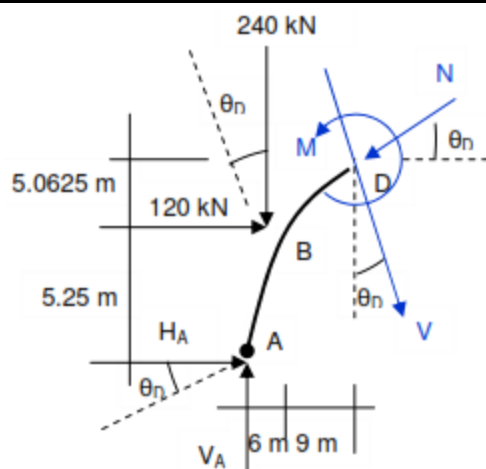
$$\tan \theta_C = dy/dx|_C = 13.5/24 \Rightarrow \theta_C = 29.36^\circ$$

Using the N-V axes system:

$$\begin{aligned} \sum F_N = N - V_A \sin \theta_C - H_A \cos \theta_C - 120 \cos \theta_C \\ + 240 \sin \theta_C = 0 \Rightarrow \mathbf{N = 406.4 \text{ kN}} \end{aligned}$$

$$\begin{aligned} \sum F_V = V - V_A \cos \theta_C + H_A \sin \theta_C + 120 \sin \theta_C \\ + 240 \cos \theta_C = 0 \Rightarrow \mathbf{V = -46.7 \text{ kN}} \end{aligned}$$

$$\begin{aligned} \sum M_C = V_A \times 10.5 - H_A \times 8.203 - 120 \times 2.953 \\ - 240 \times 4.5 - M = 0 \Rightarrow \mathbf{M = 641.9 \text{ kNm}} \end{aligned}$$

Joint to the left of D (the 150 kN will not feature)

Isolate a fbd of the arch portion A to D

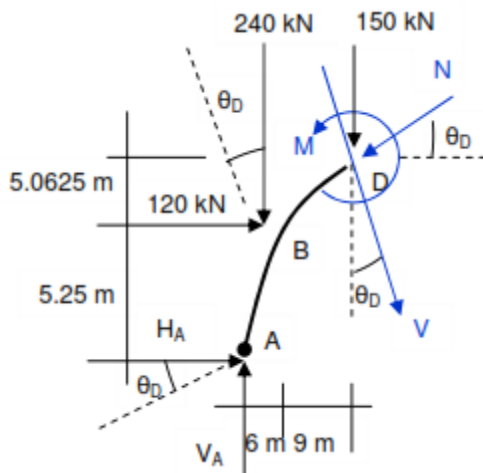
$$\tan \theta_D = dy/dx \big|_D = 9/24 \Rightarrow \theta_D = 20.56^\circ$$

$$\begin{aligned} \sum F_N = N - V_A \sin \theta_D - H_A \cos \theta_D - 120 \cos \theta_D \\ + 240 \sin \theta_D = 0 \Rightarrow \mathbf{N = 408.8 \text{ kN}} \end{aligned}$$

$$\begin{aligned} \sum F_V = V - V_A \cos \theta_D + H_A \sin \theta_D + 120 \sin \theta_D \\ + 240 \cos \theta_D = 0 \Rightarrow \mathbf{V = 16.1 \text{ kN}} \end{aligned}$$

$$\begin{aligned} \sum M_D = V_A \times 15 - H_A \times 10.3125 - 120 \times 5.0625 \\ - 240 \times 9 - M = 0 \Rightarrow \mathbf{M = 560.2 \text{ kNm}} \end{aligned}$$

Just to the right of D (the 150 kN will feature)



$$\tan \theta_D = dy/dx \big|_D = 9/24 \Rightarrow \theta_D = 20.56^\circ$$

$$\begin{aligned} \sum F_N = N - V_A \sin \theta_D - H_A \cos \theta_D - 120 \cos \theta_D \\ + 240 \sin \theta_D + 150 \sin \theta_D = 0 \Rightarrow \mathbf{N = 356.1 \text{ kN}} \end{aligned}$$

$$\begin{aligned} \sum F_V = V - V_A \cos \theta_D + H_A \sin \theta_D + 120 \sin \theta_D \\ + 240 \cos \theta_D + 150 \cos \theta_D = 0 \Rightarrow \mathbf{V = -124.4 \text{ kN}} \end{aligned}$$

$$\begin{aligned} \sum M_D = V_A \times 15 - H_A \times 10.3125 - 120 \times 5.0625 \\ - 240 \times 9 - M = 0 \Rightarrow \mathbf{M = 560.2 \text{ kNm}} \end{aligned}$$

At E

Isolate a fbd of the arch portion E to G¹⁵

¹⁵ Note that the moment calculated at E is zero, consistent with the pin at E

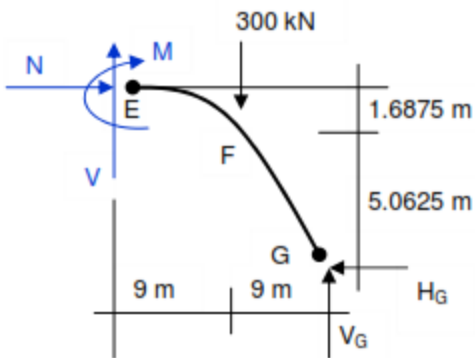
The slope $\theta_E = 0$

$$\sum F_N = N - H_G = 0 \Rightarrow N = 377.1 \text{ kN}$$

$$\sum F_V = V - 300 + V_G = 0 \Rightarrow V = 8.6 \text{ kN}$$

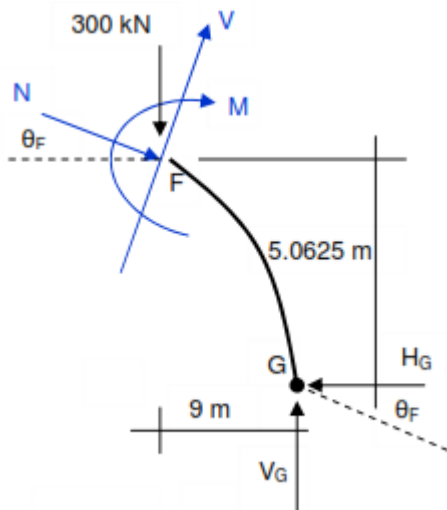
$$\sum M_E = M + 300 \times 9 + H_G \times 6.75 - V_G \times 18 = 0$$

$$\Rightarrow M = 0$$



Just to the left of F (the 300 kN will feature)

Isolate a fbd of the arch portion G to F



$$\tan \theta_F = dy/dx|_F = 9/24 \Rightarrow \theta_F = 20.56^\circ$$

$$\sum F_N = N - V_G \sin \theta_F - H_G \cos \theta_F + 300 \sin \theta_F = 0$$

$$\Rightarrow N = 350.0 \text{ kN}$$

$$\sum F_V = V + V_G \cos \theta_F - H_G \sin \theta_F - 300 \cos \theta_F = 0$$

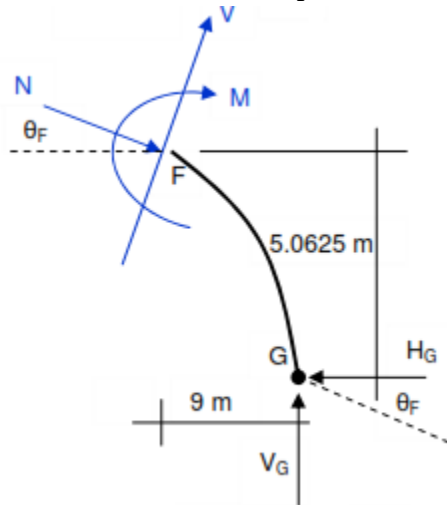
$$\Rightarrow V = 140.5 \text{ kN}$$

$$\sum M_F = -V_G \times 9 + H_G \times 5.0625 + M = 0$$

$$\Rightarrow M = 713.5 \text{ kNm}$$

Just to the right of F (the 300 kN will not feature)

Isolate a fbd of the arch portion G to F



$$\tan \theta_F = dy/dx \big|_F = 9/24 \Rightarrow \theta_F = 20.56^\circ$$

$$\sum F_N = N - V_G \sin \theta_F - H_G \cos \theta_F = 0$$

$$\Rightarrow \mathbf{N = 455.4 \text{ kN}}$$

$$\sum F_V = V + V_G \cos \theta_F - H_G \sin \theta_F = 0$$

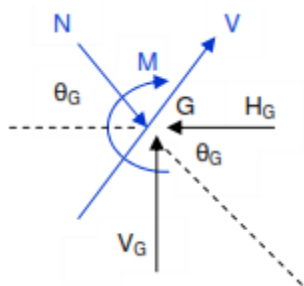
$$\Rightarrow \mathbf{V = -140.4 \text{ kN}}$$

$$\sum M_F = -V_G \times 9 + H_G \times 5.0625 + M = 0$$

$$\Rightarrow \mathbf{M = 713.5 \text{ kNm}}$$

At G

Isolate a fbd of the arch portion, *just above point G*



$$\tan \theta_G = dy/dx \big|_G = 18/24 \Rightarrow \theta_G = 36.87^\circ$$

$$\sum F_N = N - V_G \sin \theta_G - H_B \cos \theta_G = 0 \Rightarrow \mathbf{N = 476.5 \text{ kN}}$$

$$\sum F_V = V + V_G \cos \theta_G - H_B \sin \theta_G = 0 \Rightarrow \mathbf{V = -6.9 \text{ kN}}$$

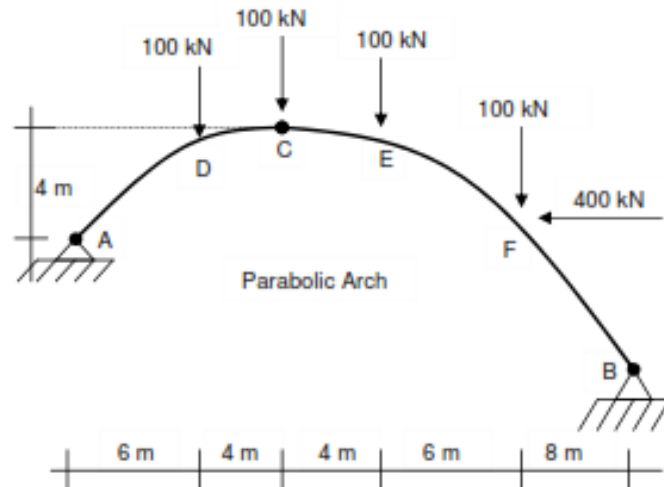
$$\sum M_G = \mathbf{M = 0}$$

TUTORIAL 5: THE ARCH

1

Determine N , V and M at D and F .

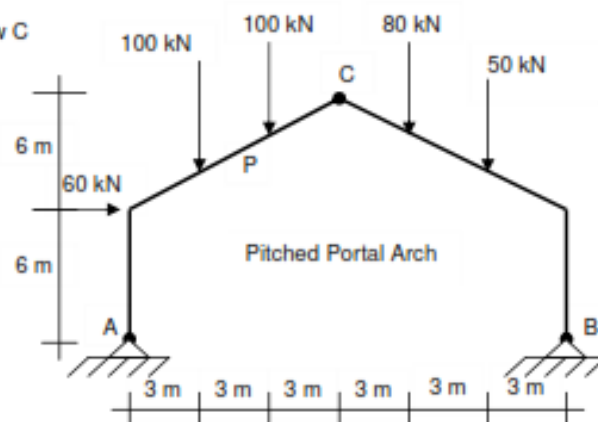
$V_A = 211.91 \text{ kN}$, $H_A = 429.76 \text{ kN}$
 $V_B = 188.09 \text{ kN}$, $H_B = -29.76 \text{ kN}$
 Just to the left of D
 $N = 473.9 \text{ kN}$, $V = 70.8 \text{ kN}$, $M = -172.2 \text{ kNm}$
 Just to the right of D
 $N = 443.4 \text{ kN}$, $V = -24.4 \text{ kN}$, $M = -172.2 \text{ kNm}$
 Just to the left of F
 $N = 390.6 \text{ kN}$, $V = 199.7 \text{ kN}$, $M = 1238.2 \text{ kNm}$
 Just to the right of F
 $N = 140.7 \text{ kN}$, $V = -128.3 \text{ kN}$, $M = 1238.2 \text{ kNm}$



2

Determine N , V and M at P which is 3 m below C

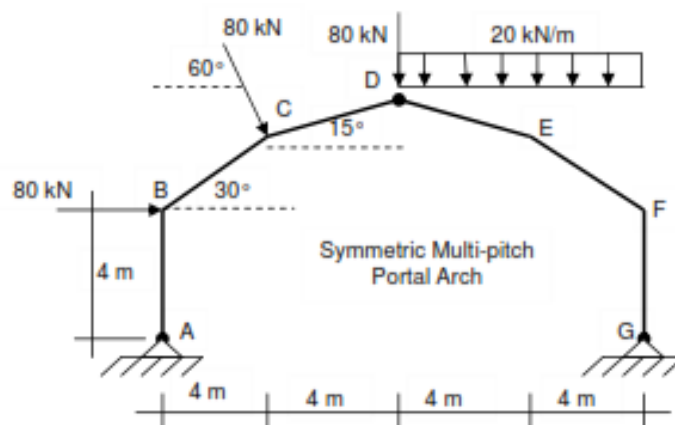
$V_A = 165 \text{ kN}$, $H_A = 18.75 \text{ kN}$
 $V_B = 165 \text{ kN}$, $H_B = -78.75 \text{ kN}$
 At P : $N = 101.58 \text{ kN}$
 $V = 10.40 \text{ kN}$
 $M = 243.75 \text{ kNm}$



3

Determine N , V and M at F .

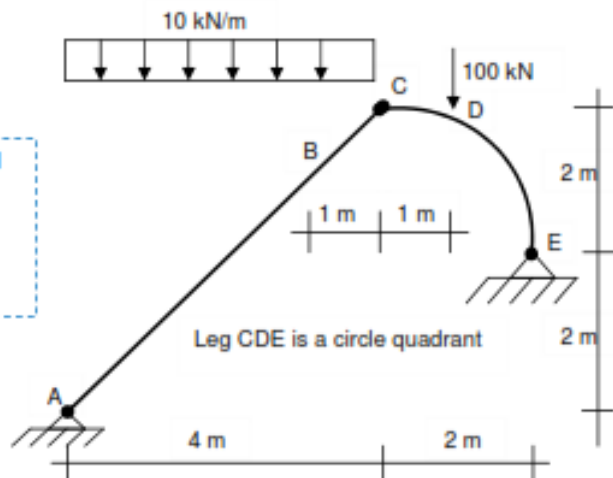
$V_A = 96.19 \text{ kN}$, $H_A = 24.25 \text{ kN}$
 $V_G = 213.09 \text{ kN}$, $H_G = 144.25 \text{ kN}$
 Just above F
 $N = 231.47 \text{ kN}$, $V = -112.42 \text{ kN}$
 $M = -577 \text{ kNm}$
 Just below F
 $N = 213.09 \text{ kN}$, $V = 144.25 \text{ kN}$
 $M = -577 \text{ kNm}$



4

Determine N , V and M (a) at point B , and
(b) just to the right of D

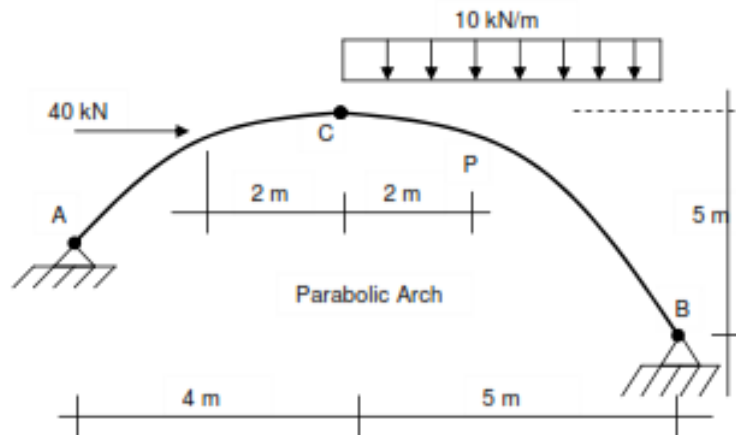
$V_A = 55 \text{ kN}$, $H_A = 35 \text{ kN}$, $V_E = 85 \text{ kN}$, $H_E = -35 \text{ kN}$
At B : $N = 42.43 \text{ kN}$, $V = -7.07 \text{ kN}$, $M = 15 \text{ kNm}$
Just to the right of D
 $N = 72.8 \text{ kN}$, $V = -56.1 \text{ kN}$, $M = 24.4 \text{ kNm}$



5

Determine N , V and M at P

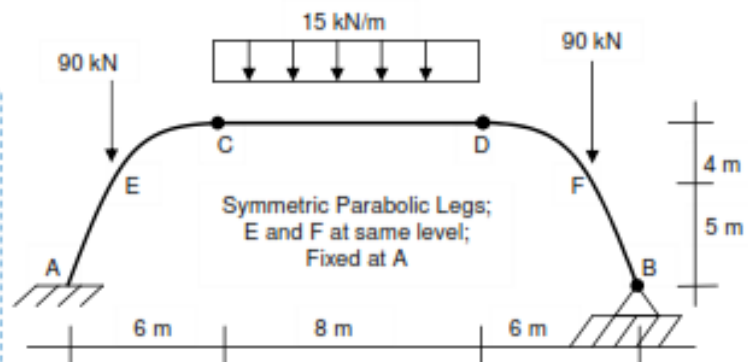
$V_A = -2.22 \text{ kN}$, $H_A = -12.78 \text{ kN}$
 $V_B = 52.22 \text{ kN}$, $H_B = -27.22 \text{ kN}$
At P : $N = 35.1 \text{ kN}$, $V = -0.35 \text{ kN}$
And $M = -2.66 \text{ kNm}$



6

Determine N , V and M at F

$V_B = 150 \text{ kN}$, $H_B = -60 \text{ kN}$
Just to the left of F
 $N = 80.5 \text{ kN}$, $V = 26.83 \text{ kN}$, $M = 0$
Just to the right of F
 $N = 160.99 \text{ kN}$, $V = -13.43 \text{ kN}$, $M = 0$



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